

LLM-POWERED ACTIVE LEARNING FOR COST-EFFECTIVE TEXT CLASSIFICATION

by

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The above committee determined that the thesis is acceptable in form and content and that a satisfactory knowledge of the field covered by the thesis was demonstrated by the candidate during an oral examination. A signed copy of the Certificate of Approval is available from the School of Graduate and Postdoctoral Studies.

Abstract

This thesis presents an LLM-powered active learning framework for cost-effective text classification, addressing the challenge of potential LLM annotation errors while balancing annotation quality and model accuracy. Our methodology combines human and large language model (LLM) annotations using uncertainty sampling and confidence scoring. Starting with a small, labeled seed set, the model iteratively selects the most informative data points for annotation, reducing labeling costs while maximizing performance. To simulate real-world scenarios, a dynamically updated proxy validation set mirrors the distribution of the unlabeled pool, enabling reliable performance estimation throughout training. The Performance Improvement Cost Ratio (PICR) is introduced as an objective stopping criterion to optimize the balance between costs and accuracy gains. Additionally, role-based prompting enhances annotation quality, creating a scalable framework adaptable to diverse text classification tasks. Experimental results demonstrate that the proposed approach achieves human-comparable performance at reduced costs, underscoring its potential for practical applications.

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would like to express my deepest gratitude to my supervisor, Professor Massoud Mahrouchi, for his invaluable guidance, encouragement, and support throughout this research journey. His insights and expertise have been instrumental in shaping this work, and his mentorship has been a constant source of motivation.

I am also grateful to my committee members for their valuable feedback and constructive suggestions, which helped enhance the quality of this thesis. Special thanks go to my colleagues and friends who shared in this journey, offering advice, encouragement, and countless discussions that enriched my understanding.

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Lastly, I extend my heartfelt thanks to my family and friends, whose unwavering support and patience have been my foundation throughout this journey. Their belief in me has been a source of strength, and I am forever grateful for their encouragement and love.

Author's Declaration

I hereby declare that this thesis consists of original work of which I have authored. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

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Hamidreza Rouzegar

Statement of Contributions

This thesis is primarily based on two papers co-authored by myself, Hamidreza Rouzegar, and my supervisor, Dr. Masoud Makrehchi. The first paper, “Enhancing Text Classification through LLM-Driven Active Learning and Human Annotation,” published in the Proceedings of The 18th Linguistic Annotation Workshop (LAW-XVIII) by the Association for Computational Linguistics (ACL), forms the foundation of this thesis. It presents our novel approach to integrating Large Language Models (LLMs) into Active Learning frameworks for text classification tasks.

The second paper, “The Role of Role Design in In-Context Learning for Large Language Models,” currently under review for the ACL Rolling Review, extends the research presented in the first paper. It explores the impact of role-based prompt design on the performance of various LLMs in zero-shot and few-shot learning scenarios. Both papers contribute significantly to all chapters of the thesis, providing the core methodology, experimental framework, and insights into optimizing LLM performance in text classification tasks.

The ideas and methodologies developed in these two papers also contributed to a third publication, “Generative AI for Enhancing Active Learning in Education: A Comparative Study of GPT-3.5 and GPT-4 in Crafting Customized Test Questions,” presented at the 37th Canadian Conference on Artificial Intelligence (CAIAC). While not directly included in this thesis, this paper demonstrates the broader applicability of our active learning approach in educational contexts.

As the primary author of all three papers, I was responsible for the conceptualization of the research, design and implementation of experiments, data analysis, and manuscript preparation. Dr. Makrehchi provided invaluable guidance throughout the research process, contributed to the refinement of ideas, and assisted in manuscript revision. All work presented in this thesis was conducted under Dr. Makrehchi’s supervision at Ontario Tech University.

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List of Abbreviations

LLM	Large Language Model
NLP	Natural Language Processing
ICL	In-Context Learning
GPT	Generative Pre-trained Transformer
IMDB	Internet Movie Database
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
CRF	Conditional Random Field
SLU	Spoken Language Understanding
AL	Active Learning
AL-SVM	Active Learning-supported SVM
ALLSH	Active Learning guided by Local Sensitivity and Hardness
AGI	Artificial General Intelligence
KATE	Knn-Augmented in-conText Example selection
EPR	Efficient Prompt Retrieval
SG-ICL	Self-generated In-context Learning
APE	Automatic Prompt Engineer
CoT	Chain-of-Thought
MoT	Memory-of-Thought
SuperICL	Super In-Context Learning
iCAP	Iterative Context-Aware Prompter
TAPP	Task-Agnostic Prefix Prompt
RAG	Retrieval Augmented Generation
SQBC	Synthetic Query-by-Committee
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
ZeroU	Zero-shot User Prompt
ZeroSU	Zero-shot System and User Prompt
FewU	Few-shot User Prompt
FewSU	Few-shot System and User Prompt
FewSUA	Few-shot System, User, and Assistant Prompt

Chapter 1

Introduction

1.1 Overview

The advancement of machine learning and deep learning has brought about significant improvements in model performance, but it has also introduced new challenges. While computational costs have increased with model complexity, a persistent and often overlooked expense in supervised learning is the acquisition of labeled data [1]. Natural Language Processing (NLP), a subfield of machine learning that focuses on the interaction between computers and human language, is no exception to this trend. Consequently, annotation remains one of the primary costs in NLP tasks.

Active Learning emerged as a strategy to address this challenge by selectively choosing instances for annotation, aiming to reduce the overall cost of data labeling [2]. This approach has shown promise in optimizing learning efficiency while minimizing the need for extensive labeled datasets. Concurrently, the field of NLP has witnessed significant advancements, leading to the emergence of Large Language Models (LLMs). These models offer a potential avenue for further reducing annotation costs by serving as alternatives to human annotators in certain tasks [3]. However, relying solely on LLMs for annotation can be problematic due to potential errors and biases inherent in these models. This

limitation underscores the need for a balanced approach that leverages the strengths of both LLMs and human expertise.

1.2 Background and Problem Statements

Active Learning has seen renewed interest in NLP, particularly since 2009-2010, coinciding with the adoption of neural models in the field [4]. This approach aims to optimize learning efficiency while reducing the need for labor-intensive labeled instances, often leading to enhanced performance metrics with less training [2]. Active Learning employs various strategies to select the most informative instances for labeling. These strategies include Query-By-Committee (QBC) [5], Expected Gradient Length (EGL) [6], and uncertainty sampling [7]. These methods have shown promise in minimizing human annotation costs while maintaining or improving model performance.

The emergence of LLMs has further expanded the possibilities in NLP tasks. Models such as GPT-3 [8] and Llama [9] have demonstrated impressive capabilities across various natural language tasks, including question answering [10], sentiment analysis [11], and text classification [12]. These models have been applied to various text annotation tasks, such as political Twitter message categorization [13], relevance and topic detection in tweets [14], and hate speech classification [15]. A key advantage of LLMs is their ability to adapt to new tasks with minimal additional training. This capability is largely due to a technique known as In-Context Learning (ICL). ICL allows LLMs to understand and perform tasks based on a few examples provided in the input prompt, without the need for extensive fine-tuning on task-specific datasets. [16]

The integration of Active Learning techniques with LLMs presents a promising avenue for enhancing text classification tasks while optimizing resource utilization. Recent studies have begun to explore this intersection, examining the potential of utilizing Active Learning in Prompt-Based Uncertainty sampling with LLMs [17]. Despite the progress in

both Active Learning and LLM applications, the integration of these approaches, particularly in balancing human and machine annotations, remains an area with opportunities for further exploration.

1.3 Research Questions

Our research aims to address several crucial Research Questions (RQ) in the field of text classification using Active Learning and LLMs:

- **RQ1:** How can we effectively combine Active Learning, human expertise, and LLM annotations to improve text classification tasks while minimizing annotation costs?
- **RQ2:** What metrics or indicators can be used to determine the optimal stopping point in active learning for text classification?
- **RQ3:** What impact does role design in prompts have on LLM performance across different text classification and question-answering tasks?

1.4 Contributions and Approach

This thesis explores the intersection of Active Learning and LLMs, proposing methodologies that combine their strengths with human expertise to enhance text classification tasks. Our research focuses on improving text classification tasks through a two-part approach. The first part involves developing and evaluating a framework that combines Active Learning, using uncertainty sampling, with annotations from both human experts and GPT-3.5. This integrated system selects between human and LLM annotators based on confidence levels derived from GPT-3.5’s outputs. We test this approach on three diverse, open-source datasets: IMDB for sentiment analysis, a dataset for identifying fake news, and another for classifying movie genres.

Our methodology employs an iterative Active Learning process where we strategically

select instances for annotation based on uncertainty sampling. In each iteration, our system decides whether to use human or LLM annotation based on confidence thresholds. We implement a novel stopping criterion using our proposed Performance Improvement Cost Ratio (PICR) metric, which helps determine the optimal point to terminate the annotation process based on the trade-off between performance gains and annotation costs.

Building upon these findings, we then extend our investigation to examine the impact of role design in zero-shot and few-shot experiments using instruction-tuned LLMs. This second phase of our research aims to enhance the performance of our initial Active Learning pipeline through optimized prompt engineering. For this part of the study, we utilize GPT-3.5 and GPT-4 from OpenAI, as well as Llama2-7b and Llama2-13b from Meta [9].

To the best of our knowledge, this study represents the first comprehensive evaluation of combining human annotators, Active Learning, and GPT-3.5 in text classification tasks, as well as the first systematic assessment of role design effectiveness in prompts across multiple datasets and LLMs. Our experimental results show that our hybrid approach achieves comparable or superior performance to traditional fully-human annotation while reducing annotation costs. Our key contributions include:

- A combined approach that reduces manual annotation costs while leveraging both human expertise and advanced machine learning techniques (RQ1)
- A proxy validation set for estimating model performance and pool quality (RQ2)
- The Performance Improvement Cost Ratio (PICR) metric for assessing annotation cost-effectiveness (RQ2)
- Insights into effective prompt design strategies for improving LLM performance in various NLP tasks (RQ3)

1.5 Thesis Structure

The remainder of this thesis is structured as follows: First, we provide a detailed background on Active Learning, LLMs, and In-Context Learning, situating our work within the broader context of NLP research. Next, we present our methodology, describing the experimental setup, including the datasets used, the Active Learning framework, and the role design configurations for LLMs. We then present our results, offering a comprehensive analysis of the performance of our proposed pipeline across different text classification tasks and LLM configurations. This is followed by a discussion of our findings, exploring the implications for future research and practical applications in NLP. Finally, we conclude with a summary of our contributions and suggestions for future work.

Chapter 2

Research Background and Related Works

2.1 Active Learning in Text Classification

Active Learning , as illustrated in the pipeline shown in Figure 2.1, is a machine learning paradigm that aims to achieve high accuracy using as few labeled training instances as possible, thereby minimizing the cost of obtaining labeled data [2]. This approach allows the learning algorithm to interactively query the user (or some other information source) to label new data points with the desired outputs [4]. By strategically selecting the most informative instances for annotation, Active Learning can significantly reduce the amount of labeled data required for training while maintaining or even improving model performance.

In the context of pool-based active learning, the learner selects the most informative instances for labeling from a pool of unlabeled data. Following the notation used by Settles and Craven [18], The generic pool-based algorithm can be described as presented in Algorithm 1. In this algorithmic framework, a function $\phi(\cdot)$ is employed to evaluate the informativeness of an unlabeled instance. It should be noted that the definition of this

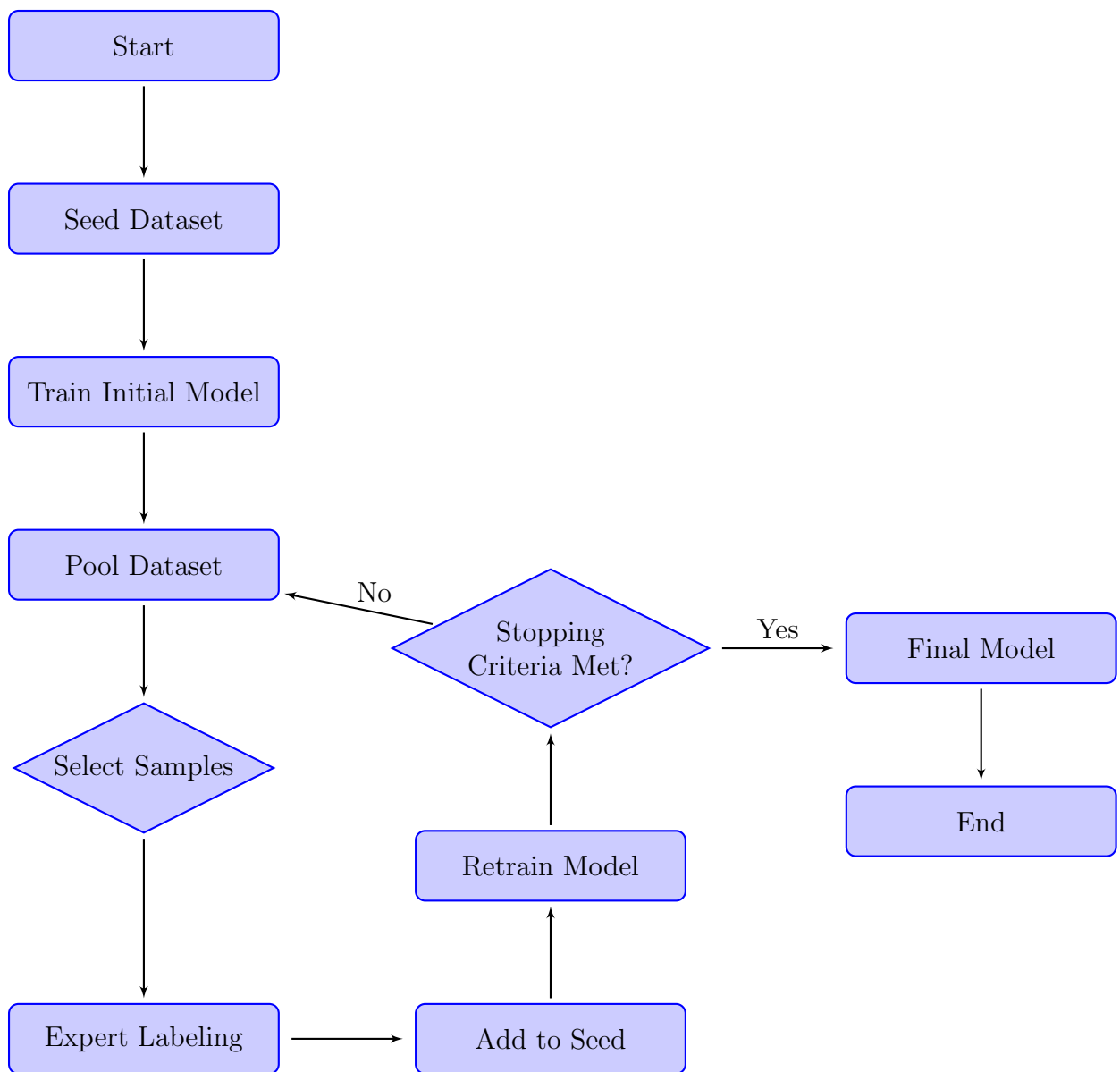


Figure 2.1: Active Learning Pipeline.

function is considered central to the majority of active learning methods, with various approaches being proposed to quantify informativeness.

Algorithm 1 Pool-based Active Learning [18]

Given: Labeled set $\mathcal{L}$, unlabeled pool $\mathcal{U}$, query strategy $\phi(\cdot)$, query batch size $\mathcal{B}$
repeat
 $\theta = \text{train}(\mathcal{L})$
 for $b = 1$ to $\mathcal{B}$ **do**
 $x_b^* = \arg \max_{x \in \mathcal{U}} \phi(x)$
 $\mathcal{L} = \mathcal{L} \cup \langle x_b^*, \text{label}(x_b^*) \rangle$
 $\mathcal{U} = \mathcal{U} - x_b^*$
until some stopping criterion

There are several strategies for defining $\phi(\cdot)$, each with its own rationale and characteristics. The first and most common approach is Uncertainty Sampling, [7] which selects instances that the model is least certain about, aiming to clarify decision boundaries. This method is particularly effective in text classification tasks and forms the primary focus of our research.

Another popular strategy is Query-By-Committee (QBC), [5] which uses multiple models to vote on the labeling of instances, selecting those with the most disagreement among the committee. This approach can be especially useful when ensemble methods are employed in the classification task.

The third method is Expected Gradient Length (EGL), [6] which chooses instances that would result in the largest change to the current model if labeled. This strategy aims to select examples that will have the most significant impact on the model’s parameters.

Lastly, the Information Density method [19] considers both uncertainty and representativeness, selecting instances that are not only uncertain but also similar to many other instances in the dataset. This approach can be particularly useful when dealing with datasets that have distinct clusters or subgroups.

While each of these strategies has its merits, our research focuses primarily on uncertainty sampling due to its widespread use in active learning literature and its relative

simplicity in implementation, as described by Settles and Craven [18]. Uncertainty sampling includes two particularly effective methods:

- Least Confident (LC) method: This approach selects the instance for which the model’s most likely prediction is least confident. It can be formalized as:

$$\phi_{LC}(x) = 1 - P(y^*|x; \theta) \quad (2.1)$$

where y^* represents the most probable label sequence predicted by the model.

- Sequence Entropy (SE) method: This strategy measures the informativeness of an instance based on the entropy across different possible labelings. It is defined as:

$$\phi_{SE}(x) = - \sum_{\hat{y}} P(\hat{y}|x; \theta) \log P(\hat{y}|x; \theta) \quad (2.2)$$

where $\hat{y}$ range over the set of all possible labelings for x .

These uncertainty sampling methods provide a way to quantify the model’s uncertainty about an instance, allowing the active learning algorithm to prioritize the labeling of the most informative examples. By focusing on instances where the model is least confident or where there is high entropy in the predictions, these approaches aim to efficiently improve the model’s performance with minimal labeling effort.

Our research employs a variant of the Least Confident method, adapted for multi-class classification tasks. This approach allows us to quantify uncertainty based on the model’s confidence in its predictions, selecting instances where the model is least sure about its classification. The specific implementation details of this method, including the choice of classification algorithm and probability estimation technique, will be discussed in depth in the methodology section.

Various researchers have explored the application of Active Learning in text classification tasks, employing different models and strategies to optimize the balance between

annotation effort and model performance. For instance, Anderson et al. [20] explored Active Learning based on uncertainty across various models, including support vector machines, logistic regression, and decision trees. They proposed a criterion for manual annotation that identifies a certain percentage of the most uncertain predictions for each model type as a stopping point for annotation. For example, they suggested a 12.71% threshold for logistic regression to achieve desired model performance.

Exploring uncertainty-based sampling techniques, Goudjil et al. [21] applied Active Learning using support vector machines and Active Learning-supported SVM (AL-SVM) models. They introduced a thresholding mechanism, such as setting a 70% threshold for AL-SVM, to select instances with less than 70% probability for annotation. This approach demonstrates the potential of uncertainty-based sampling in reducing annotation costs while maintaining model performance.

Building on these foundations, more sophisticated Active Learning techniques have emerged. Yuan et al. [22] explored a range of Active Learning techniques, including methods based on information theory concepts. They explored Processing Surprisal, which measures the unexpectedness of data points, and Entropy, which quantifies the uncertainty in predictions. Their work provided valuable insights into the effectiveness of various selection strategies. However, one limitation of their study was the lack of a clear stopping criterion to determine the optimal number of Active Learning iterations. This gap highlights the ongoing challenge in Active Learning research of balancing annotation effort with performance gains.

Addressing some of these challenges, Zhang et al. [23] proposed an innovative method that incorporates local sensitivity and adversarial perturbations into the Active Learning loop. Their approach, named ALLSH (Active Learning guided by Local Sensitivity and Hardness), aims to mitigate the sampling bias that often results from selecting only the most uncertain examples. By using data augmentation and adversarial perturbation techniques, they were able to measure the local sensitivity of instances more accurately,

allowing them to select examples that lie near the decision boundary. This method not only improves the diversity of selected samples but also enhances the model’s robustness to minor perturbations in the input space.

Furthermore, Zhang et al. [23] introduced a “learning hardness” criterion to their selection process. This additional metric helps to identify and avoid examples that are either too difficult to learn or potentially mislabeled, which can negatively impact model performance if included in the training set. By combining these various strategies, their method achieved superior performance compared to traditional uncertainty-based sampling techniques.

Our research builds upon these foundations by introducing a novel approach that integrates Active Learning with LLMs for text classification tasks. We propose a framework that leverages the uncertainty estimates provided by LLMs to guide the selection of instances for human annotation, thereby optimizing the trade-off between annotation cost and model performance.

2.2 Active Learning with Language Models

Large Language Models (LLMs) have emerged as a transformative force in Artificial Intelligence, demonstrating remarkable versatility across a wide array of tasks [24, 25, 26]. Systems like ChatGPT [27] and GPT-4 [28] have shown capabilities that extend beyond traditional AI boundaries, leading to their increasing adoption in various domains.

As LLMs continue to evolve, there is a growing need to develop new evaluation protocols and application methodologies that can adequately assess and leverage their capabilities. One particularly promising area of research is the integration of LLMs with Active Learning frameworks, especially in the context of text classification tasks. This combination has the potential to address some of the longstanding challenges in machine learning, such as the need for large labeled datasets and the high costs associated with

expert annotation.

The integration of LLMs into Active Learning frameworks represents a significant advancement in the field. Recent research has explored various approaches to leveraging the strengths of LLMs in Active Learning scenarios. Yue Yu et al. [17] introduced PATRON, a method that incorporates prompt-based techniques for cold-start data selection in Active Learning. Their approach uses estimated uncertainty for data points and adapts two key design strategies: uncertainty propagation and a partition-then-rewrite (PTR) strategy. These strategies ensure a balance between informativeness and diversity in sample selection, addressing common challenges in Active Learning such as the cold-start problem and the need for diverse, representative samples.

Building on this foundation, Zhang et al. [29] proposed a framework where LLMs serve as active annotators. Their work underscores the efficiency of LLMs in reducing annotation costs while maintaining high accuracy. This approach demonstrates the potential of using LLMs not just as predictive models, but as integral components in the annotation process itself.

In line with these approaches, Wang [30] proposed a consistency-based strategy to identify potentially mislabeled samples, which are then annotated by humans in Active Learning settings. This “mixed annotation strategy” demonstrated that models trained with this method achieve similar or better results compared to those with human-only annotations, highlighting the cost efficiency of integrating LLMs as annotators.

The concept of integrating human annotators with machine-generated explanations was further explored in the “Beyond Labels” study. This research demonstrated the potential for more informative annotations in low-resource settings by combining human expertise with the natural language generation capabilities of LLMs. This hybrid approach suggests a promising direction for leveraging both human and machine intelligence in annotation tasks.

Additionally, Xiao et al. [31] introduced FreeAL, a collaborative learning approach

that reduces the need for human annotation by leveraging LLMs as weak annotators. This framework indicates the potential of LLMs to enhance unsupervised performance in various NLP tasks. Similarly, Bayer et al. [32] introduced ActiveLLM, an active learning approach leveraging LLMs like GPT-4 for selecting instances. This method significantly improves classification performance in few-shot scenarios and helps other Active Learning strategies overcome the cold start problem.

Furthermore, Liang et al. [33] addressed the challenge of generalized category discovery by integrating LLM feedback into an Active Learning framework. Their method employs uncertainty propagation to select high-uncertainty samples for labeling using a comparison-based prompting scheme. Kholodna et al. [34] focused on low-resource languages, proposing the integration of LLMs in the Active Learning loop for data annotation. Their evaluations with GPT-4-Turbo show significant cost savings and near-state-of-the-art performance.

Despite these advancements, Lu et al. [35] presented empirical evidence suggesting that smaller models trained with expert annotations can outperform LLMs in domain-specific tasks. This finding underscores the irreplaceable value of human expertise in certain scenarios and highlights the importance of considering the trade-offs between LLM-based and human-based annotation strategies. Similarly, Xu et al. [36] proposed ActiveRAG, combining Retrieval Augmented Generation (RAG) with Active Learning to improve question-answering tasks by developing a deeper understanding of external knowledge.

Wagner et al. [37] proposed SQBC, a new Active Learning method based on the “Query-by-Committee” approach using LLM-generated synthetic data. This method identifies the most informative unlabelled samples for manual labeling, improving stance detection performance in online political discussions. Their approach, along with other innovative strategies, illustrates the broad potential to improve NLP tasks using advanced Active Learning techniques.

Vysogorets and Gopal [38] introduced a strategy that uses pre-trained representations of LLM within the Aactive Learning loop. Using this representation, the process of acquiring labeled data is expedited and the need to retrain massive models at each iteration is minimized. This method shows that fine-tuning on the acquired data can achieve performance comparable to continuously fine-tuning, but with significantly reduced computational costs.

These recent advancements in combining Active Learning with LLMs have opened up new avenues for research in text classification and other NLP tasks. They demonstrate the potential to create more efficient, accurate, and adaptable systems by leveraging the strengths of both traditional Active Learning techniques and the powerful capabilities of modern LLMs.

2.3 Prompt Engineering and In-Context Learning

As LLMs have become increasingly powerful and versatile, the importance of effective prompt engineering and in-context learning has grown significantly. These techniques allow researchers and practitioners to leverage the vast knowledge encoded in LLMs without the need for fine-tuning, leading to more efficient and flexible NLP solutions.

Prompt engineering refers to the process of designing and optimizing input prompts to elicit desired outputs from LLMs. This involves crafting instructions, providing context, and structuring queries in ways that maximize the model’s performance on specific tasks [39, 40]. One powerful method within prompt engineering is In-Context Learning (ICL), also known as few-shot learning in the context of LLMs. [41]

ICL, also known as few-shot learning in the context of LLMs, enables models to perform tasks based on a small number of examples provided in the prompt. This approach has gained traction due to its ability to adapt LLMs to new tasks quickly and without the need for extensive labeled datasets or model retraining.

Several methods have been developed to enhance the selection and retrieval of examples for in-context learning. Liu et al. [42] introduced KATE (Knn-Augmented in-context Example selection), which uses k-nearest neighbors to choose relevant examples within the context. This method improves the quality of selected examples, leading to better task performance. Similarly, Rubin et al. [43] proposed EPR (Efficient Prompt Retrieval), providing an efficient method for interacting with and learning from LLMs through optimized prompt retrieval.

An innovative approach to reducing dependence on external demonstrations is Self-generated In-context Learning (SG-ICL), introduced by Kim et al. [44]. This method allows the LLM to generate its own examples, potentially leading to more tailored and effective in-context learning. Additionally, Sorensen et al. [45] explored the use of Mutual Information (MI) as a metric for example selection, applying concepts from information theory to quantify the shared information between variables and improve the selection of few-shot examples.

Gonen et al. [46] proposed designing prompts based on Perplexity Estimation, hypothesizing that a prompt’s effectiveness is linked to the model’s familiarity with its language. This approach offers a new perspective on prompt design, focusing on the model’s inherent language understanding.

Active Example Selection has been explored using reinforcement learning (RL) techniques. Zhang et al. [47] treated the selection process as a Markov Decision Process (MDP) and applied Q-learning to optimize example choices. Li and Qiu [48] further refined this process with their LENS (fiLter-thEN-Search) method, which filters datasets to find informative examples and then iteratively searches for the best combinations.

Wang et al. [49] proposed an approach inspired by topic models, selecting optimal demonstrations from annotated data using smaller language models and then generalizing these demonstrations to larger LLMs for ICL. The Unified Demonstration Retriever (UDR) [50] uses a multi-task listwise ranking training framework with an iterative mining

strategy to identify high-quality candidates for demonstrations across various tasks.

Instruction Induction, proposed by Honovich et al. [51], allows models to generate natural language instructions from input-output pairs, enhancing their ability to understand and follow directions. This concept was further developed in the Automatic Prompt Engineer (APE) by Zhou et al. [52], which treats instructions as programs and optimizes them by searching through a pool of candidates proposed by an LLM. The SELF-INSTRUCT framework [53] improves the instruction-following capabilities of pre-trained language models by leveraging their own generated examples.

Chain-of-Thought (CoT) prompting, introduced by Wei et al. [54], provides reasoning demonstrations that allow models to develop reasoning abilities naturally when given sufficiently complex tasks. Variants like complex CoT [55] and Auto-CoT [56] further refine these demonstrations under different conditions.

The Self-Ask method [57] enhances performance by decomposing complex questions into simpler sub-questions, which the model answers sequentially. The Memory-of-Thought (MoT) approach [58] involves pre-thinking on an unlabeled dataset and storing high-confidence thoughts as external memory, which the model can recall during testing to aid reasoning.

Super In-Context Learning (SuperICL) [59] combines LLMs with locally fine-tuned smaller models to improve performance on supervised tasks. These smaller models act as plug-ins, providing specific task knowledge while the larger models focus on general language understanding.

Iterative Context-Aware Prompter (iCAP) [60] explores an iterative prompting framework that progressively elicits relevant knowledge from LLMs for multi-step inference. Similarly, the LEAST-TO-MOST Prompting method breaks down complex problems into a series of simpler subproblems and solves them in sequence [61].

Lastly, the Task-Agnostic Prefix Prompt (TAPP) [62] is a fixed prompt prepended to the beginning of every input, regardless of the target task, facilitating zero-shot general-

ization.

These advancements in prompt engineering and in-context learning have significantly expanded the capabilities of LLMs in various NLP tasks. They provide a foundation for more efficient and effective use of LLMs, paving the way for novel applications and improvements in areas such as text classification, question answering, and other complex language understanding tasks.

Chapter 3

Methodology

In machine learning, a significant challenge is obtaining labeled data, with annotation often being a major cost in developing models. The trade-off between model performance and costs is important. Active Learning can help to reduce annotation costs while aiming to maintain good performance. Using Large Language Models (LLMs) as annotators could further reduce costs compared to human annotation, but this introduces potential errors in the annotations.

This section outlines our methodology, which combines Active Learning with both LLM and human annotations. We aim to improve performance while keeping costs low. We describe the Active Learning process, the implementation of a proxy-validation set, and the use of GPT-3.5 for data annotation and confidence scoring. Lastly, we explore role-based prompt designs to potentially improve LLM responses and the annotation process, which could enhance overall classification performance.

3.1 Active Learning Framework

Our Active Learning approach centered on uncertainty sampling. In many machine learning classification tasks, models can predict not only the class label but also the probability of their classifications for new instances. These probability scores, often

obtained through methods like softmax, indicate how confident the model is in assigning a specific class to an instance. In Active Learning, we leverage these probability scores as a measure of uncertainty or confidence for each sample in the unlabeled data pool. By ranking data points based on these predicted probabilities, we can identify and select the most uncertain instances for labeling. In our implementation, we chose to use logistic regression for its simplicity and its ability to provide probability scores, which allowed us to easily apply this uncertainty sampling strategy in our pipeline.

In each Active Learning iteration, the data samples with the highest uncertainty scores from the pool of unlabeled data are selected, adding these data samples to the training set for the next iteration. This process of focusing on high-information gain data allowed the model to learn from its previous mistakes and incrementally improve its overall performance with each iteration. By iteratively selecting and incorporating informative data, the Active Learning approach optimizes the learning process efficiently. This Active Learning approach often achieves similar performance to conventional supervised learning methods while using fewer labeled samples.

We initiated the Active Learning process by creating an initial set of data that incorporated a small fraction of the entire data available for the classification task. This set served as the starting point for the learning process, enabling the model to make its initial predictions. The rest of the data, not included in the initial set, constituted the unlabeled pool. This pool continuously provided a source of uncertain samples to be selected and labeled during each iteration of the Active Learning process.

As described in the related work section, a function $\phi(\cdot)$ evaluates the informativeness of unlabeled instances in active learning. Our method employs uncertainty sampling, utilizing the model’s predicted probabilities as a measure of uncertainty. We implement this using the `predict_proba` function from scikit-learn’s logistic regression model [63], which serves as our $\phi(\cdot)$ function.

Recall that the generic pool-based active learning algorithm is described in Algorithm

1. The `predict_proba` function in scikit-learn’s `LogisticRegression` class handles both binary and multi-class scenarios [64], estimating probabilities for each class and quantifying uncertainty across various classification tasks.

1. For binary classification or when using one-vs-rest approach: The function calculates the probability of each class using the logistic function.
2. For multi-class problems with multinomial option: The function uses the softmax function to compute probabilities.

We adapt the previously discussed $\phi_{LC}(x)$ function (Equation 2.1) proposed by Settles and Craven [18] to our scenario. The uncertainty measure $\phi(x)$ for an instance x is then calculated as:

$$\phi(x) = 1 - \max_i P(y_i|x) \quad (3.1)$$

Where $P(y_i|x)$ is the probability estimate for class i , obtained from model prediction probabilities (using `predict_proba`). Here, $\max_i$ represents the probability associated with the predicted class by our model, indicating the model’s highest confidence level for class i given input x . In our implementation, we used the following steps:

1. Train the logistic regression model on the current labeled dataset.
2. Apply the model’s predicted probability to all instances in the unlabeled pool.
3. Calculate the uncertainty using the equation above.
4. Rank the unlabeled instances based on their uncertainty scores.
5. Select the top N most uncertain instances for labeling.

In multi-class problems, `predict_proba` returns an array with dimensions (`n_samples`, `n_classes`). Each row in this array represents a sample, and each column represents the probability estimate for a specific class. For these multi-class scenarios, the softmax function is used to compute these probabilities:

$$P(y_i|x) = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \quad (3.2)$$

where z_i are the raw scores (logits) for each class, and k is the number of classes.

This approach allows us to efficiently identify the most informative instances for labeling across both binary and multi-class problems, maximizing the information gain from each newly labeled example. By focusing on the instances where the model is least certain, we can refine the decision boundaries more effectively, leading to improved classification performance with fewer labeled examples compared to random sampling methods.

3.2 Proxy-Validation Set

One of our key contributions is the creation of a proxy-validation set, addressing a significant challenge in real-world Active Learning scenarios. When dealing with a pool of unlabeled data, it’s typically impossible to know the true performance of our model in labeling the pool. This uncertainty makes it difficult to determine when to stop the iterative annotation process. To address this, we propose using a subset of our initially labeled data as a proxy-validation set to estimate the performance of the model at each iteration.

As shown in Algorithm 2, the proxy-validation set serves as a representative sample of labeled data, emulating the statistical distribution of the main unlabeled pool. During each Active Learning iteration, we compute the performance of the model on this set. To maintain consistency, we apply the same confidence threshold for data removal to both the proxy-validation set and the main unlabeled pool. The remaining data in the proxy-validation set provides an estimation of the accuracy of the main pool.

By utilizing this proxy-validation approach, we can make informed decisions about the optimal point to stop adding more annotated data. This allows us to balance the

Algorithm 2 Proxy Validation

Given: Total dataset D , labeled set L , unlabeled set U , proxy-validation set $P_t \subset L$, uncertainty threshold θ_t , iteration t

Define: $\phi(x) = 1 - \max_i P(y_i|x)$ where $P(y_i|x)$ is the probability estimate for class i
for each iteration t **do**

$U_t \leftarrow$ remaining unlabeled pool

Calculate $\phi(x)$ for all $x \in P_t$

$P_{t+1} \leftarrow x \in P_t : \phi(x) > \theta_t$

Calculate $F1(P_t)$

Estimate $F1(U_t) \approx F1(P_t)$

Update proxy set statistics

Output: Updated P_{t+1} , estimated $F1(U_t)$

trade-off between model performance and annotation costs, even in scenarios where the true labels of the unlabeled pool are unknown. The proxy-validation set serves as a reliable indicator for estimating the model’s performance and correlates with the overall pool quality.

The implementation of the proxy-validation set involved the following steps:

1. Randomly select a subset of the total data to serve as the proxy-validation set.
2. The goal is for this subset to maintain a similar distribution of classes as the main dataset
3. Apply the same uncertainty sampling and low-confidence removal processes to this set as to the main unlabeled pool.
4. Use the performance on this set to gauge overall model performance and guide decisions about when to terminate the Active Learning process.

Let D be the total dataset, $L \subset D$ be the labeled part, and $U \subset D$ be the unlabeled part. As the active learning process progresses: $T_t \subset L$ is the training set, $P_t \subset L$ is the proxy-validation set, and $U_t \subset U$ is the remaining unlabeled pool at iteration t . We can express this as:

$$D = L \cup U \tag{3.3}$$

$$L = T_t \cup P_t \tag{3.4}$$

As iterations progress, instances move from U to L , specifically into T_t , while P_t is initially created from a subset of L and evolves alongside the active learning process.

We calculate uncertainty based on Equation 3.1, which applies to both binary and multi-class classification tasks. Consider an example where U_t contains instances {A, B, C, D} with probability scores $\{(0.2, 0.8), (0.48, 0.52), (0.57, 0.43), (0.6, 0.4)\}$. For each instance, ϕ is calculated by subtracting the highest predicted class probability from 1, where the highest probability.

The ϕ values for each instance are as follows:

- Instance A: $1 - 0.8 = 0.2$
- Instance B: $1 - 0.52 = 0.48$
- Instance C: $1 - 0.57 = 0.43$
- Instance D: $1 - 0.6 = 0.4$

If $k = 2$ (instances to be removed), we select the two most uncertain instances (B and C), and set the threshold $\theta_t = 0.43$. We propose to select instances from U_t to move to T_t based on this uncertainty:

$$S_t = x \in U_t : \phi(x) \leq \theta_t \tag{3.5}$$

where $|S_t| = k$. For the proxy-validation set P_t , we apply a similar process. If P_t contains instances {E, F, G} with probability scores $\{(0.51, 0.49), (0.9, 0.1), (0.15, 0.85)\}$, we calculate the ϕ values as follows:

- Instance E: $1 - 0.51 = 0.49$
- Instance F: $1 - 0.9 = 0.1$
- Instance G: $1 - 0.85 = 0.15$

Using the threshold $\theta_t = 0.43$, we retain instances F and G for the next iteration and remove instance E using the following equation:

$$P_{t+1} = x \in P_t : \phi(x) > \theta_t \quad (3.6)$$

This demonstrates how the proxy-validation set simulates uncertainty reduction in the main pool without contributing data to the training set. We propose using the F1 score of the model on P_t at iteration t , denoted as $F1_P(t)$, to estimate the model performance on the entire dataset:

$$F1(P_t) \approx F1(U_t) \quad (3.7)$$

where $F1_U(t)$ is the true (but unknown) F1 score on the entire unlabeled pool.

To validate the effectiveness of our proxy-validation set, we designed evaluation framework. This framework aims to quantify the relationship between the F1 scores of the proxy-validation set and the real pool over multiple iterations of the Active Learning process. We selected the following metrics for this evaluation:

1. Correlation Coefficients:

- Pearson Correlation Coefficient: This measures the linear correlation between the proxy and pool F1 scores. A value close to 1 indicates a strong positive correlation.
- Spearman's Rank Correlation: This assesses monotonic relationships, whether linear or not. It's particularly useful if the relationship might not be strictly linear.

2. Error Metrics:

- Mean Absolute Error (MAE): This measures the average absolute difference between the proxy and pool F1 scores, providing an easily interpretable measure of the typical discrepancy.

- Root Mean Square Error (RMSE): This gives the standard deviation of the residuals (prediction errors), penalizing larger errors more heavily than MAE.

3. Coefficient of Determination:

- R-squared (R^2): This indicates the proportion of the variance in the pool F1 scores that is predictable from the proxy F1 scores. It provides a measure of how well the proxy scores explain the variation in the pool scores.

These metrics were chosen to provide a comprehensive view of the relationship between the proxy-validation set and the real pool. The correlation coefficients assess the strength and direction of the relationship, while the error metrics quantify the typical magnitude of discrepancies between the proxy and pool scores. The R-squared value provides an overall measure of how well the proxy scores can predict the pool scores. By combining these different metrics, we can evaluate not only the strength of the relationship between the proxy and pool scores but also the magnitude and consistency of any differences. This multi-faceted approach allows for a robust assessment of the proxy-validation set’s reliability as an indicator of overall pool quality.

The evaluation framework is designed to be applied across different datasets and variations of our method, allowing us to assess the consistency and robustness of the proxy-validation set’s performance. This comprehensive evaluation provides a solid foundation for determining the reliability of using the proxy-validation set to guide decisions in our Active Learning process, such as when to terminate the learning iterations.

3.3 LLM-based Data Annotation

Building upon our Active Learning framework, we integrated GPT-3.5 into our data annotation process to enhance efficiency and scalability across our diverse datasets. Our approach extends beyond simple classification, incorporating confidence assessments from the LLM to inform our sampling strategy and annotation decisions.

For each dataset, which will be discussed in detail in the experimental setup section, we crafted specialized prompts that instruct GPT-3.5 to provide both a classification label and a confidence score. This dual output allows us to capture not just the model’s decision, but also its certainty.

The confidence scores obtained from GPT-3.5 play a crucial role in our methodology. We use these scores to dynamically determine when to rely on LLM annotations and when to seek human input. This adaptive approach allows us to optimize the balance between automated and manual annotation, potentially reducing the overall annotation cost while maintaining high data quality.

To further refine our LLM-based annotation, we implemented a few-shot learning technique. In cases where GPT-3.5 initial confidence falls below a predetermined threshold, we provide the model with a small set of labeled examples before requesting a new classification. This method aims to improve the performance of the model on challenging instances without resorting to human annotation, effectively creating a middle ground between fully automated and human-dependent labeling.

Our methodology also explores the impact of varying confidence thresholds on the mixing of LLM and human annotations. By adjusting these thresholds, we can investigate how different levels of LLM certainty affect the overall quality of our dataset and the subsequent performance of our classification models. This approach allows us to tailor our annotation strategy to the specific characteristics of each dataset and classification task.

This adaptive approach allows us to optimize the balance between automated and manual annotation, potentially reducing overall annotation cost while maintaining high data quality. It effectively replicates the essence of Active Learning, where the model iteratively improves by focusing on the most informative or uncertain samples.

The integration of LLM-based annotation into our Active Learning framework presents both opportunities and challenges. While it offers the potential for rapid, cost-effective

labeling, it also introduces the risk of propagating errors inherent in the LLM’s training data. To address this, our methodology incorporates continuous evaluation of the LLM’s annotations against a subset of human-labeled data, allowing us to monitor and mitigate any systematic errors that may emerge.

Furthermore, our approach considers the potential for LLM annotations to influence the direction of the Active Learning process itself. By carefully tracking how LLM-annotated data affects the selection of subsequent instances for labeling, we can gain insights into the interplay between automated annotation and the trajectory of model improvement in an Active Learning context.

To gain a comprehensive understanding of the effects of LLM integration, we designed our experiments to compare models trained on datasets with varying degrees of LLM involvement in the annotation process. This ranges from fully human-annotated datasets to those relying heavily on LLM annotations, with several hybrid approaches in between. By analyzing the performance differences across these models, we aim to identify optimal strategies for leveraging LLMs in Active Learning pipelines for different types of classification tasks.

To evaluate the effectiveness of our uncertainty-based Active Learning approach, we established a baseline comparison using random selection. For each experiment conducted with our proposed uncertainty sampling method, we ran a parallel experiment using random sampling instead. This random selection approach serves as a control, allowing us to isolate and quantify the benefits of our uncertainty-based method.

In the random selection baseline, instead of choosing instances based on the model uncertainty, we randomly select the same number of instances from the unlabeled pool at each iteration. This process mimics a scenario where no intelligent selection strategy is employed, providing a fair comparison to demonstrate the advantages of our Active Learning approach.

By comparing the performance of models trained using uncertainty sampling against

those trained with random sampling, we can assess the efficiency gains and improvements in classification accuracy attributable to our Active Learning strategy. This comparison is crucial for validating the effectiveness of our approach across different datasets and experimental conditions.

Furthermore, this baseline comparison helps us understand the learning dynamics of our models under different data selection strategies. It allows us to analyze how quickly our uncertainty-based approach converges to optimal performance compared to random selection.

Through this multifaceted approach, our research seeks to improve the efficiency of the data labeling process and contribute to the broader understanding of how large language models can be effectively integrated into Active Learning methodologies. Our goal is to develop annotation strategies that are adaptable, efficient, and capable of producing high-quality labeled datasets across a variety of NLP tasks, while maintaining the core principles of Active Learning within the unique context of working with LLMs.

3.4 Cost-Effectiveness Analysis and Stopping Criteria

To optimize the Active Learning process, we developed a metric called the Performance Improvement Cost Ratio (PICR) to assess the cost-effectiveness of continued annotation. This metric aims to balance the trade-off between performance improvement and annotation costs, providing a data-driven approach to determine when to stop the Active Learning iterations. Our method calculates the cost required to achieve a specified percentage improvement in performance, regardless of the number of iterations. This approach allows us to evaluate the marginal benefit of additional annotations in terms of both performance gain and associated costs. The general calculation process can be

formalized as:

$$\text{PICR} = \frac{\text{Cost}_i - \text{Cost}_j}{(\text{Performance}_i - \text{Performance}_j) * (100/\alpha)} \quad (3.8)$$

where i represents the current iteration, j represents the last iteration with a significant performance improvement, and α is the desired percentage improvement threshold (e.g., 1%, 0.5%, 2%, etc.). In our study, we used F1 score as our performance metric and set $\alpha = 1$, calculating the cost for each 1% improvement in F1 score.

This metric provides insights into the diminishing returns of the annotation process. As we progress through iterations, we generally observe that achieving performance improvements becomes increasingly costly. This trend can be visualized as a series of horizontal and vertical lines, where longer horizontal lines indicate periods of annotation without significant performance gains, and steeper vertical lines represent higher costs for smaller improvements, which will be discussed in the Results section.

Using this visualization, we can identify potential stopping points in the Active Learning process. For instance, a prolonged horizontal line suggests a phase where additional data is not yielding performance improvements, indicating a possible stopping point. We can formalize this by setting a threshold, such as halting the process if we do not see a 1% increase in performance over 10 iterations.

Moreover, this approach allows us to set cost-based thresholds. For example, we might decide that spending \$1000 for a 1% performance improvement is acceptable, but \$1500 for the same gain is not cost-effective. This flexible framework enables us to tailor our stopping criteria to specific budget constraints and performance goals.

As shown in the Algorithm 3, by analyzing this cost-effectiveness metric throughout the Active Learning iterations, we can make informed decisions about when to terminate the process. This method ensures that we optimize not just for performance, but for the efficient use of annotation resources, providing a more holistic approach to Active Learning optimization that considers both the diminishing returns in performance and the escalating costs of annotation.

This approach goes beyond traditional stopping criteria that focus solely on performance metrics or fixed iteration counts. Instead, it integrates the practical consideration of annotation costs, which is crucial for real-world applications of Active Learning, allowing for a nuanced, data-driven decision-making process in managing the trade-off between annotation effort and model performance improvement.

Algorithm 3 Active Learning Pipeline with PICR-based Stopping Criteria

Given: Seed dataset $\mathcal{S}$, pool dataset $\mathcal{P}$, GPT confidence threshold τ , stopping criterion based on PICR

repeat

Step 1: Train Initial Model

 Train model θ using seed dataset $\mathcal{S}$

Step 2: Sample Selection

 Select samples $\mathcal{X} \subset \mathcal{P}$ based on query strategy

Step 3: Expert Labeling

for each $x_i \in \mathcal{X}$ **do**

if $\text{GPTConfidence}(x_i) > \tau$ **then**

 Annotate x_i with GPT-generated labels

else

 Annotate x_i with human-generated labels

$\mathcal{S} = \mathcal{S} \cup \{x_i\}$

 ▷ Add newly labeled samples to seed

Step 4: Retrain Model

 Retrain model θ on the updated seed dataset $\mathcal{S}$

Step 5: Calculate PICR

 Compute PICR as:

$$\text{PICR} = \frac{\text{Cost}_i - \text{Cost}_j}{(\text{Performance}_i - \text{Performance}_j) \cdot \left(\frac{100}{\alpha}\right)}$$

until PICR-based stopping criterion is met

Output: Final model θ

3.5 Role Design in Prompts

A central focus of our research, alongside balancing cost and performance, has been improving the performance of LLM annotations themselves. In this phase of our study, we explored the impact of role design in prompts as a means to enhance LLM perfor-

mance in text classification tasks. This investigation aims to demonstrate how carefully structured prompts can significantly improve the quality of LLM-generated annotations. Such improvements could pave the way for even greater reliance on LLM annotations and further reduce the need for human involvement in our Active Learning framework.

Our approach to role design involves structuring prompts with distinct roles to provide a more contextualized and effective input to the model. By optimizing these prompt structures, we seek to improve the accuracy and reliability of LLM outputs. The roles we considered are:

1. System: Provides high-level instructions that guide the overall task and set the context for the interaction.
2. User: Presents specific queries or prompts that the model needs to respond to, effectively driving the interaction and simulating real-world usage scenarios.
3. Assistant: Represents the model response to the user’s queries, which should adhere to the system instructions and accurately address the user’s prompts.

We developed a range of prompt configurations that vary in their structure and the amount of context provided, spanning from zero-shot approaches with minimal context to few-shot approaches that include examples to guide the model understanding. The designs incorporate different combinations of these role distinctions within the prompts. For this phase of our research, we focused on a subset of our original datasets: the IMDB dataset for sentiment analysis and the Movie Genres dataset for genre classification. To demonstrate the robustness of our approach across different task types, we also included two question-answering datasets. This selection allows us to assess the generalizability of our role design approach across diverse classification and question-answering scenarios.

It’s important to note that while our initial Active Learning experiments used GPT-3.5 exclusively due to its balance of performance and accessibility, our role design experiments incorporate a wider range of models. These include GPT-3.5 and GPT-4o from

OpenAI, as well as Llama2-7b and Llama2-13b from Meta. This decision was driven by budget constraints and our primary focus on demonstrating the effectiveness of our proposed methodologies rather than exhaustively comparing model performances.

The computational demands of using open-source models like Llama for the initial confidence-based selection in our Active Learning pipeline precluded their inclusion in that phase of the research. However, their inclusion in the role design experiments allows us to explore the effectiveness of our approach across different model architectures and sizes.

Our methodology involved systematically testing these various prompt designs across the selected datasets and models. We evaluated the performance of each prompt design in terms of classification or question-answering accuracy and the model’s adherence to the specified output format.

Through this investigation of role design in prompts, we seek to contribute to the growing body of knowledge on effective prompt engineering for LLMs. Our goal is to potentially offer guidelines for structuring prompts to maximize performance across various NLP tasks, thereby enhancing the effectiveness of LLMs in practical applications.

The specific details of each prompt configuration and examples of their implementation will be provided in the experimental setup section of this thesis.

3.6 Evaluation Metrics

Our study employs a range of evaluation metrics to provide a comprehensive assessment of our approaches across different experimental conditions. These metrics are designed to evaluate both the performance and efficiency of our methods.

F1 Score: The F1 score serves as our primary evaluation metric for measuring the accuracy of classifications and answers produced by our models. This metric provides a balanced measure of precision and recall, offering a single score that reflects the model’s

effectiveness across different classes and datasets.

Structural Accuracy: Introduced in our role design experiments, structural accuracy measures how closely the model’s output adheres to the requested structure in the prompt. For example: - In movie genre classification, the output should be a single word among the specified genres. - For sentiment analysis, the output should be a single word, either “positive” or “negative.” - In QA tasks, the answer should be a single capital letter among the choices (A, B, C, or D). We examine structural accuracy regardless of the correctness of the answer, focusing on the model’s ability to follow instructions and produce outputs in the desired format.

Cost Estimation: To evaluate the economic feasibility and efficiency of our approach, we conducted a thorough cost estimation analysis for each experimental condition. The costs were computed based on two primary sources:

a) Human Annotation Costs: We used the pricing structure provided by the AI Platform Data Labeling Service for human labels.¹ The cost was calculated based on units per human labeler, with each unit encompassing a fixed number of words. The total cost of human annotation was determined by multiplying the total number of units labeled by human annotators with the per-unit price.

b) LLM Annotation Costs: For machine-generated labels, we utilized the pricing structure for the GPT-3.5 API, which charges based on the number of tokens processed. The total cost of machine annotation was calculated by multiplying the total number of tokens processed by GPT-3.5 with the cost per token.²

For each of our experiments, which will be discussed in the experimental setup section, we computed the price at each iteration. We added the human annotation cost and the LLM annotation cost, depending on how much of our data came from true labels and how much from GPT annotations. It’s important to note that our dataset was already

¹<https://cloud.google.com/ai-platform/data-labeling/pricing>

²<https://openai.com/pricing>

labeled, and we used the human labeling cost structure as an estimation based on the number of words. We assumed we were using human labelers for this cost calculation, although we already had the labels. In contrast, the GPT-3.5 API cost represents the actual expense we incurred for annotation using the model.

The total cost at each iteration t can be represented by the following formula:

$$C_{T_t} = n \cdot C_H + m \cdot C_{L_t} \quad (3.9)$$

Where C_{T_t} is the total cost at iteration t , n is the number of instances in the training set with true labels, C_H is the cost per instance for human annotation, m is the number of LLM-annotated instances, and C_{L_t} is the cost of LLM annotation per instance at iteration t .

Chapter 4

Experimental Results

Our experimental setup is designed to comprehensively evaluate the performance of Large Language Models (LLMs) in data annotation for text classification tasks, with a particular focus on GPT-3.5. The experiments span two main phases, each addressing distinct aspects of our research questions.

In the first phase, we explore the impact of using GPT-3.5 for data annotation under various scenarios, including different confidence thresholds and combinations of human and GPT-3.5 annotations. This phase focuses on three diverse datasets: IMDB movie reviews for sentiment analysis [65], Movie Genres for multi-class classification [66], and a fake news dataset for binary classification. [67] The diversity in these datasets allows us to assess the generalizability of our approach across different text classification tasks and data characteristics.

The second phase of our experiments extends our investigation to examine the impact of role design in prompts on the performance of various LLMs. For this phase, we expanded our model selection to include GPT-3.5-turbo-0125 and GPT-4o from OpenAI, as well as Llama2-7b-chat and Llama2-13b-chat from Meta. This broader selection enables us to explore how different model architectures and sizes respond to various prompt designs. In addition to the IMDB and Movie Genres datasets used in the first phase, we

introduced two question-answering datasets (commonsense qa [68] and ai2 arc [69]) to demonstrate the robustness of our role design approach across different task types.

It’s important to note that our use of GPT-3.5 exclusively for the initial Active Learning experiments was driven by its balance of performance and accessibility. The inclusion of a wider range of models in our role design experiments was motivated by our primary focus on demonstrating the effectiveness of our proposed methodologies rather than exhaustively comparing model performances. Additionally, the computational demands of using open-source models like Llama for the initial confidence-based selection in our Active Learning pipeline precluded their inclusion in that phase of the research.

Throughout our experiments, we maintain a consistent focus on optimizing the balance between annotation accuracy and cost-effectiveness. This is achieved through careful manipulation of confidence thresholds, a strategic combination of human and machine annotations, and exploration of various prompt designs. Our goal is to provide insights that can guide the development of more efficient and accurate text classification pipelines, combining human expertise and state-of-the-art language models.

In the following sections, we will detail our datasets, experimental scenarios, evaluation metrics, and analytical procedures. We will also discuss the limitations and considerations of our experimental setup, providing a comprehensive view of our research methodology.

4.1 Datasets and Preprocessing

Our experiments utilize a diverse set of datasets, carefully selected to evaluate the effectiveness of our proposed methodologies across various text classification tasks and question-answering scenarios. For all datasets, we created balanced subsets to ensure fair evaluation and prevent bias in our experiments.

1. IMDB Movie Reviews Dataset [65] This dataset consists of movie reviews labeled as

positive or negative, making it ideal for binary sentiment classification. It allows us to evaluate our methods’ effectiveness in discerning sentiment from user-generated content, a common task in many real-world applications such as product reviews or social media analysis.

2. Fake News Dataset [67] This dataset comprises news articles, where the task is to classify each article as either fake or real. This binary classification challenge represents a critical application in the current digital landscape, where distinguishing between genuine and false information is increasingly important.
3. Movie Genres Dataset [66] For multi-class classification, we employed a Movie Genres dataset. We selected the four largest classes - comedy, drama, action, and horror - and created a balanced subset. The task involves classifying movie plots into these genre categories, representing a more complex classification challenge that mirrors real-world content categorization scenarios.
4. CommonsenseQA Dataset [68] This is a multiple-choice question-answering dataset that requires various types of commonsense knowledge to predict the correct answers. Each question has five possible answers (A, B, C, D, E), testing the model’s ability to reason and apply common sense in diverse contexts.
5. AI2 Reasoning Challenge Dataset [69] This dataset consists of genuine grade-school level, multiple-choice science questions, designed to encourage research in advanced question-answering. It’s partitioned into a Challenge Set and an Easy Set, with the Challenge Set containing only questions answered incorrectly by both retrieval-based and word co-occurrence algorithms.

Our preprocessing approach differed between the two phases of our experiment:

For Phase 1 (Active Learning with GPT-3.5), we implemented a comprehensive preprocessing strategy. We began by cleaning the text data, which involved removing symbols and converting all text to lowercase. Our data splitting approach was structured

as follows: we used 2% of the whole dataset as the initial seed and allocated 5% of the dataset for the proxy validation set. The active learning process consisted of 250 iterations, with each iteration adding 0.2% of the initial dataset size. This methodical approach resulted in our experiments testing from 2% to 52% of the total data across all three datasets (IMDB, Fake News, and Movie Genres).

In Phase 2 (Role Design Experiments), we adopted a different preprocessing approach. We minimized text alterations, refraining from removing symbols or modifying the text structure. This decision was made to preserve the original content and to take full advantage of the LLMs’ ability to process raw text. For the few-shot prompts, we exercised great care in selecting examples. We ensured that these examples had no overlap with the test data, a crucial step in preventing data leakage and maintaining the integrity of our experiments.

Our careful data preparation and experimental design allow us to rigorously test the effectiveness of our Active Learning approach with GPT-3.5 in the first phase, and then explore the impact of role design in prompts across different LLMs in the second phase. This two-phase approach provides a nuanced understanding of how LLMs can be effectively leveraged in various NLP tasks.

4.2 Experimental Setups

The experimental setup is organized into two phases: the first focuses on Active Learning with GPT-3.5, while the second explores Role Design Experiments with multiple LLMs.

4.2.1 Phase 1: Active Learning with GPT-3.5

In the first phase, we focused on leveraging GPT-3.5 for data annotation in an Active Learning framework. We explored various scenarios to understand the impact of different annotation strategies on model performance and cost-effectiveness.

GPT-3.5 Labels Only: The first scenario we tested was using GPT-3.5 labels exclusively. This allowed us to assess the capability of the LLM in handling the annotation task independently, providing insights into its strengths and limitations when used as a standalone annotation tool.

Human Labels Only: As a baseline and point of comparison, we also conducted experiments using human labels only. This traditional approach to data annotation served as a control, allowing us to measure the effectiveness of GPT-3.5 annotations against the gold standard of human expertise. The comparison between these two scenarios provides valuable insights into the potential of LLMs to augment or potentially replace human annotation in certain contexts.

Hybrid Labels based on Confidence Threshold: A key focus of our research was exploring hybrid approaches that combine the strengths of both GPT-3.5 and human annotators. To this end, we conducted a series of experiments with varying confidence thresholds. In these hybrid setups, we utilized GPT-3.5 confidence scores to determine when to use its labels and when to defer to human annotation. We tested three specific confidence thresholds: 90%, 80%, and 70%.

In the 90% confidence threshold experiment, we only accepted GPT-3.5 labels when its confidence score exceeded 90%, relying on human annotations for all other cases. This high threshold aimed to maximize the accuracy of machine annotations while still leveraging GPT-3.5 capabilities for the most clear-cut cases. The 80% and 70% threshold experiments followed a similar pattern but with increasingly lenient criteria for accepting GPT-3.5 labels. These varying thresholds allowed us to explore the trade-offs between increased automation and potential drops in annotation accuracy.

GPT-3.5 with Few-Shot: To further investigate the potential of GPT-3.5, we also experimented with few-shot learning in an Active Learning context. This approach was particularly interesting for data points where GPT-3.5 initially showed low confidence. Unlike the hybrid approach that combines human and GPT-3.5 annotations, this setup

utilizes GPT-3.5 few-shot learning capabilities exclusively. We implemented a strategy where data points above a certain confidence threshold (70% for IMDB, 80% for Movie Genres and Fake News) were handled with zero-shot learning, while those below the threshold were processed with few-shot learning using three examples. This adaptive approach aimed to enhance GPT-3.5 performance on more challenging instances without resorting to human annotation.

These thresholds were chosen based on an analysis of our datasets, which revealed that lower thresholds did not significantly change many labels, while higher thresholds led to an excessively large portion of the data being re-annotated. With these thresholds, we managed to target approximately 10 to 15 percent of the IMDB and Movie Genres datasets and about 4 percent of the fake news dataset, ensuring a manageable yet effective scope for applying few-shot learning.

Baseline Comparison (Random Data Addition): To rigorously evaluate the effectiveness of our uncertainty-based Active Learning approach, we established a baseline comparison using random selection. For each experiment conducted with our proposed uncertainty sampling method, we ran a parallel experiment using random sampling instead. This random selection approach serves as a control, allowing us to isolate and quantify the benefits of our uncertainty-based method.

In the random selection baseline, instead of choosing instances based on the model’s uncertainty, we randomly select the same number of instances from the unlabeled pool at each iteration. This process mimics a scenario where no intelligent selection strategy is employed, providing a fair comparison to demonstrate the advantages of our Active Learning approach.

By comparing the performance of models trained using uncertainty sampling against those trained with random sampling, we can assess the efficiency gains and improvements in classification accuracy attributable to our Active Learning strategy. This comparison is crucial for validating the effectiveness of our approach across different datasets and

experimental scenarios.

4.2.2 Phase 2: Role Design Experiments with Multiple LLMs

Building upon our primary objective of enhancing performance while minimizing costs in text classification tasks, the second phase of our study explored an avenue for further refinement of our active learning pipeline. Specifically, we investigated the impact of role design in prompts on the performance of Large Language Models (LLMs) in zero-shot and few-shot learning scenarios. This exploration aimed to demonstrate how we could potentially increase the effectiveness of LLMs themselves, a crucial component of our pipeline, independent of the active learning process.

For this phase, we utilized four state-of-the-art LLMs: GPT-3.5-turbo-0125 and GPT-4o from OpenAI, and Llama2-7b-chat and Llama2-13b-chat from Meta. These models were selected for their robust performance and widespread applicability in various NLP tasks. While we didn't incorporate these specific scenarios into our active learning pipeline due to model availability changes and budget constraints, this investigation opens up possibilities for future enhancements.

To maintain continuity with our earlier work, we chose two datasets from our previous experiments to demonstrate the potential improvements achievable through this design. Furthermore, our investigation into various scenarios in this phase aligns with a key finding from our main active learning pipeline: the correlation between the accuracy of labels and overall pipeline performance. Just as we observed that more correct labels led to better performance in our main pipeline, we anticipate that enhancing the accuracy of LLM-generated labels through improved prompt design could similarly boost the overall effectiveness of our system. This parallel underscores the potential synergy between optimizing LLM performance and refining our active learning approach.

To investigate the impact of role design, we experimented with the following prompt configurations. Importantly, the main content of the prompt remained the same across all

configurations; the key difference lay in how the prompt was split into distinct roles. This approach allowed us to isolate the effect of role-based structuring on model performance, which is the primary focus and contribution of our study.

As discussed earlier in the Methodology chapter on role design, the “system” provides high-level instructions that guide the overall task and set the context for the interaction. This includes specifying the format, the rules for the responses, or any additional context necessary for the task. The “user” presents specific queries or prompts that the model needs to respond to, effectively driving the interaction and simulating real-world usage scenarios. The “assistant” is the model’s response to the user’s queries, which should adhere to the system’s instructions and accurately address the user’s prompts. We experimented with five different prompt designs:

1. Zero-shot User Prompt (ZeroU): In this configuration, the entire prompt is provided as a user instruction without any role distinctions. This serves as our baseline for zero-shot learning.
2. Zero-shot System and User Prompt (ZeroSU): Here, we split the prompt into two parts. The initial part is provided as a system instruction, setting the context and rules, followed by the user input containing the specific query or task.
3. Few-shot User Prompt (FewU): This design incorporates multiple examples of questions and answers within the user prompt. It aims to provide the model with context through examples, all within the user’s role.
4. Few-shot System and User Prompt (FewSU): In this configuration, we provide examples with clear distinctions between system and user roles. The system role provides the overall context and rules, while the user role contains both the examples and the query.
5. Few-shot System, User, and Assistant Prompt (FewSUA): This is our most comprehensive design, including examples with system instructions, user inputs, and

assistant responses. It aims to provide the model with a complete interaction pattern to follow.

Each of these designs was carefully constructed to maintain consistency in the main content while varying the role structure. This allowed us to isolate and analyze the impact of role-based prompting on model performance across different tasks and datasets.

By systematically testing these prompt designs across our selected datasets and models, we aim to provide insights into optimal strategies for structuring prompts to maximize LLM performance in various NLP tasks. Our goal is to contribute to the growing body of knowledge on effective prompt engineering for LLMs.

4.2.3 Resource Utilization and Configuration

Our experiments were conducted using a combination of local systems and high-performance computing resources, tailored to the requirements of different models.

For the Active Learning experiments with GPT-3.5, we utilized the OpenAI API on our local system. The API configuration was crucial for maintaining consistency across experiments. Specifically, the temperature parameter, which controls the randomness of the model’s output, was set to 0. In language models, temperature typically ranges from 0 to 1, where 0 produces the most predictable output and 1 the most diverse. By setting the temperature to 0, we ensured more deterministic responses, which was essential for our comparative analysis.

For the Llama models, we leveraged the computational resources provided by Alliance Canada servers. The Llama-7B model was run on a single GPU with 64GB memory, which provided sufficient computational power for this smaller model. For the larger Llama-13B model, we utilized two parallel GPUs, each with 64GB memory, to accommodate its increased size and computational requirements. This parallel configuration allowed for efficient processing of the larger model without compromising on speed or performance.

For reproducibility purposes, detailed information about our experimental setup, including code repositories, specific model configurations are available on GitHub.^{1 2}

4.2.4 Proxy Validation Implementation

The implementation of the proxy validation set in our experimental setup builds upon the core concepts outlined in the methodology. Our approach involves a dynamic updating process that mirrors the uncertainty reduction in the main pool, ensuring that the proxy set maintains a consistent statistical relationship with the overall dataset throughout the Active Learning iterations. As it showed in Algorithm 2, we implemented this process as follows:

1. At each iteration, we calculate uncertainty scores for all instances in the proxy validation set using the same $\phi(x)$ function applied to the main pool. This involves computing predicted probabilities using the current model and applying the uncertainty measure.
2. We then apply the uncertainty threshold θ_t , determined from the main pool selection process, to the proxy set. This step ensures consistency between the main pool and proxy set selection criteria.
3. The proxy validation set is updated by removing instances with uncertainty below θ_t . This mimics the sample selection process in the main pool without actually contributing these samples to the training set.
4. After updating the proxy set, we calculate the F1 score on the remaining instances, which serves as our estimate for the model’s performance on the entire unlabeled pool.

To evaluate the effectiveness of our proxy validation approach, we implemented a

¹GitHub Code Phase 1

²GitHub Code Phase 2

comprehensive statistical analysis framework. At each iteration of the Active Learning process, we compute the following metrics to quantify the relationship between the F1 scores of the proxy validation set and the real pool:

- Pearson correlation coefficient and its p-value
- Spearman’s rank correlation coefficient and its p-value
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R-squared (R^2) value

These metrics are calculated by comparing the F1 score obtained from the proxy validation set with the F1 score computed on a pool set with true labels. This comparison allows us to assess how well the proxy validation set estimates the model’s true performance at each iteration.

By tracking these metrics throughout the Active Learning process, we can observe how the reliability of the proxy validation set evolves over time and under different experimental scenarios. This analysis provides insights into the robustness of our proxy validation approach and its effectiveness in guiding the Active Learning process without relying on extensive labeled data.

4.2.5 Cost-Effectiveness Analysis and Stopping Criteria

To accurately assess the economic implications of our annotation strategies, we developed a comprehensive cost estimation approach. For each experiment, we calculated the cumulative cost at each iteration, combining both human annotation costs and LLM annotation costs. The human annotation cost was estimated based on a standard pricing structure for data labeling services³, calculated according to the number of words in each instance. Although our datasets were pre-labeled, we used this estimation to simulate

³<https://cloud.google.com/ai-platform/data-labeling/pricing>

real-world annotation expenses. In contrast, the GPT-3.5 API cost ⁴ represented the actual expenditure incurred for model-based annotations. This dual approach allowed us to compare the financial implications of human versus machine annotations realistically.

Building upon this cost calculation framework, we developed a metric to assess the cost-effectiveness of continued annotation, balancing performance improvement against annotation costs. This metric calculates the cost required to achieve each 1% improvement in F1 score, allowing us to evaluate the marginal benefit of additional annotations in terms of both performance gain and associated costs.

This approach provides insights into the diminishing returns of the annotation process. As we progress through iterations, we generally observe that achieving performance improvements becomes increasingly costly. This trend can be visualized as a series of horizontal and vertical lines, where longer horizontal lines indicate periods of annotation without significant performance gains, and steeper vertical lines represent higher costs for smaller improvements.

Using this metric, we can identify potential stopping points in the Active Learning process. For instance, a prolonged horizontal line suggests a phase where additional data isn't yielding performance improvements, indicating a possible stopping point. We can formalize this by setting a threshold, such as halting the process if we don't see a 1% increase in performance over a certain number of iterations.

Algorithm 4 Stopping Criterion for Active Learning

Require: PICR, threshold, window

- 1: **if** length of PICR < window **then**
 - 2: **return** False
 - 3: **if** last value in PICR > threshold **then**
 - 4: **return** False
 - 5: **return** True
-

Moreover, this approach allows us to set cost-based thresholds. For example, we might decide that spending \$1000 for a 1% performance improvement is acceptable, but

⁴<https://openai.com/pricing>

\$1500 for the same gain is not cost-effective.

This Algorithm 4 checks if the average cost-effectiveness over the last window iterations exceeds a predefined threshold. By analyzing this cost-effectiveness metric throughout the Active Learning iterations, we can make informed decisions about when to terminate the process.

Chapter 5

Analysis of Results

Our study on integrating Large Language Models (LLMs) into Active Learning frameworks for text classification tasks yielded insightful results across three distinct datasets: IMDB Reviews, Fake News, and Movie Genres. Each dataset presented unique challenges, allowing us to assess the versatility and effectiveness of our proposed methodologies.

Our findings reveal significant improvements in classification accuracy and cost-effectiveness when combining LLMs with Active Learning techniques. This results section is organized into four main parts: (1) quantitative results comparing various annotation methods, (2) visualized results illustrating performance trends, (3) analysis of our novel proxy validation approach, and (4) insights from our role design experiments with different LLM prompts. Through this comprehensive analysis, we demonstrate the potential of our approach to revolutionize text classification tasks across diverse domains.

5.1 Quantitative Analysis of Results

As shown in Table 5.1, the IMDB dataset, a binary classification task for sentiment analysis, revealed a consistent improvement in F1 scores across all annotation methods as the data portion increased from 10% to 50%. The GPT-only approach demonstrated competitive performance, achieving an F1 score of 0.9629 at 50% data, compared to

Portion	10%		20%		30%		40%		50%	
Method	Cost	F1	Cost	F1	Cost	F1	Cost	F1	Cost	F1
GPT only	0.46	0.8201	0.92	0.8651	1.3809	0.9152	1.84	0.9439	2.30	0.9629
GPT conf > 70	74.44	0.8548	170.45	0.9077	283.37	0.9446	373.04	0.9671	442.42	0.978
GPT conf > 80	219.04	0.8522	476.13	0.908	733.22	0.9469	958.60	0.9696	1151.85	0.9802
GPT conf > 90	369.12	0.8533	750.09	0.9112	1139.51	0.9495	1517.10	0.9700	1873.55	0.9800
Human only	423.24	0.8597	846.49	0.9085	1269.73	0.9475	1692.98	0.9693	2116.22	0.9796
Few-shot < 70	0.95	0.8469	1.77	0.8973	2.60	0.9407	3.33	0.9631	4.07	0.9773

Table 5.1: IMDB Dataset Results: F1 Score and Costs (in USD) for Various Annotation Methods on the IMDB Dataset.

Portion	10%		20%		30%		40%		50%	
Method	Cost	F1	Cost	F1	Cost	F1	Cost	F1	Cost	F1
GPT only	1.53	0.8099	3.06	0.8339	4.59	0.8576	6.12	0.8837	7.66	0.9041
GPT conf > 70	7.16	0.8099	12.91	0.8394	25.70	0.8573	34.27	0.8764	41.44	0.9028
GPT conf > 80	39.53	0.7889	77.66	0.8319	97.49	0.8675	159.54	0.9017	187.82	0.9243
GPT conf > 90	437.85	0.9007	843.33	0.9426	1289.63	0.9646	1714.82	0.9771	2156.90	0.9871
Human only	1409.02	0.9074	2818.04	0.956	4227.07	0.9701	5636.09	0.9745	7045.11	0.9791
Few-shot < 70	1.59	0.7838	3.14	0.8335	4.69	0.8667	6.24	0.8888	7.81	0.9092

Table 5.2: Fake News Dataset Results: Comparative Analysis of F1 Scores and Annotation Costs (in USD) for the Fake News Dataset.

0.9796 for human-only annotations. Notably, the GPT-only method accomplished this at a fraction of the cost, ranging from \$0.46 to \$2.30, in stark contrast to the \$423.24 to \$2116.22 range for human-only annotations.

The hybrid approaches, which combined GPT annotations with human oversight based on varying confidence thresholds, showcased an interesting trade-off between cost and performance. As evident in Table 5.1, higher confidence thresholds led to improved F1 scores but also increased costs. Notably, the hybrid method with a 90% confidence threshold achieved an F1 score of almost 0.98 at 50% data, matching human-only performance while reducing costs by approximately 11.5% (\$1873.55 vs. \$2116.22).

A particularly promising approach emerged in the form of few-shot learning for data points with confidence below 70%. This method achieved F1 scores comparable to higher-cost methods (0.9773 at 50% data) while maintaining costs similar to the GPT-only

approach (\$4.07 vs. \$2.30). This finding suggests that strategic application of few-shot learning can significantly enhance the efficiency of LLM-based annotation without substantially increasing costs.

The Fake News dataset, representing a binary classification task for distinguishing genuine from false information, presented a more challenging scenario compared to the IMDB dataset. As illustrated in Table 5.2, we observed a wider range of F1 scores across methods, indicating the increased complexity of this classification task.

The GPT-only method demonstrated moderate performance improvement as the data portion increased, with F1 scores rising from 0.8099 at 10% data to 0.9041 at 50% data. Notably, this improvement was achieved at minimal cost, ranging from \$1.53 to \$7.66. However, the human-only annotations consistently outperformed all other methods, achieving the highest F1 scores ranging from 0.9074 at 10% data to 0.9791 at 50% data. This superior performance came at substantially higher costs, from \$1409.02 to \$7045.11.

Interestingly, the hybrid method with a 90% confidence threshold emerged as a promising compromise between performance and cost. This approach achieved F1 scores comparable to human-only annotations at 50% data while reducing costs by approximately 69% (\$2156.90 vs. \$7045.11). This finding suggests that for complex tasks like fake news detection, leveraging high-confidence LLM annotations can significantly reduce costs without compromising on accuracy.

The few-shot learning approach for data points with confidence below 80% showed modest improvements over the GPT-only method, achieving an F1 score of 0.9092 at 50% data compared to 0.9041 for GPT-only. However, its performance still lagged behind the higher-confidence hybrid methods and human-only annotations. This indicates that for tasks with higher complexity or potential societal impact, like fake news detection, more sophisticated strategies or higher confidence thresholds may be necessary to achieve optimal performance.

Portion	10%		20%		30%		40%		50%	
Method	Cost	F1	Cost	F1	Cost	F1	Cost	F1	Cost	F1
GPT only	0.81	0.3530	1.62	0.4066	2.43	0.4236	3.24	0.4233	4.05	0.4337
GPT conf > 70	6.01	0.3767	8.31	0.4056	17.31	0.4235	25.56	0.4308	31.58	0.4364
GPT conf > 80	125.82	0.2688	169.79	0.4806	233.10	0.4998	299.39	0.5053	347.83	0.5358
GPT conf > 90	263.48	0.6212	534.40	0.716	809.79	0.7513	1078.48	0.7959	1353.87	0.8271
Human only	744.92	0.6039	1489.84	0.711	2234.77	0.7520	2979.69	0.8099	3724.61	0.8443
Few-shot < 70	0.88	0.3700	1.74	0.4223	2.60	0.4726	3.4615	0.4761	4.31	0.4996

Table 5.3: Movie Genres Dataset Results: F1 Scores and Costs (in USD) Across Different Annotation Methods for the Movie Genres Dataset.

The Movie Genres dataset, representing a multi-class classification task, presented unique challenges that highlighted the limitations and strengths of different annotation methods. As shown in Table 5.3, F1 scores were generally lower across all methods compared to the binary classification tasks, underscoring the increased complexity of multi-class classification.

The GPT-only method showed limited performance on this task, with F1 scores ranging from 0.3530 at 10% data to 0.4337 at 50% data. This modest improvement came at minimal costs, from \$0.81 to \$4.05. However, the relatively low F1 scores suggest that GPT-3.5, when used in isolation, struggles with more complex multi-class classification tasks without additional support or fine-tuning.

Human-only annotations achieved the highest F1 scores, ranging from 0.6039 at 10% data to 0.8443 at 50% data. While this performance was superior, it came at significant costs, from \$744.92 to \$3724.61. This stark contrast between human and GPT-only performance highlights the challenges LLMs face in complex classification tasks and the continued value of human expertise in such scenarios.

Interestingly, the hybrid method with a 90% confidence threshold showed substantial improvement over lower thresholds and the GPT-only approach. This method achieved F1 scores close to human-only performance (0.8271 vs. 0.8443 at 50% data) at about 36% of the cost (\$1353.87 vs. \$3724.61). This finding suggests that for multi-class

tasks, leveraging high-confidence LLM annotations can significantly reduce costs while maintaining performance close to human-level accuracy.

The few-shot learning approach for data points with confidence below 80% showed modest improvements over the GPT-only method, achieving an F1 score of 0.4996 at 50% data compared to 0.4337 for GPT-only. However, its performance still lagged considerably behind the higher-confidence hybrid methods and human-only annotations. This indicates that for complex multi-class tasks, more sophisticated few-shot learning strategies or higher confidence thresholds may be necessary to bridge the performance gap.

Our experiments across the IMDB, Fake News, and Movie Genres datasets revealed several key trends and insights:

- Across all datasets and annotation methods, we observed a consistent improvement in F1 scores as the data portion increased from 10% to 50%. This trend underscores the importance of data quantity in enhancing classification performance, regardless of the annotation method used.
- The performance of GPT-only annotations varied significantly based on task complexity. For the binary sentiment analysis task (IMDB), GPT-3.5 performed remarkably well, nearly matching human-level accuracy. However, its performance declined for the more nuanced fake news detection task and struggled considerably with the multi-class movie genre classification. This highlights the need to consider task complexity when deciding on an annotation strategy.
- In all datasets, LLM-based methods (GPT-only and hybrid approaches) demonstrated significant cost savings compared to human-only annotations. The magnitude of these savings varied by task, with the most substantial reductions observed in the IMDB dataset.
- Hybrid methods, particularly those with high confidence thresholds (90%), con-

sistently outperformed GPT-only annotations and often approached or matched human-level accuracy. This was especially evident in the more complex Fake News and Movie Genres tasks, where the hybrid approach balanced performance and cost effectively.

- The few-shot learning approach showed promise, particularly in the IMDB dataset, where it achieved high accuracy at costs similar to GPT-only annotations. However, its effectiveness was less pronounced in more complex tasks, suggesting that the utility of few-shot learning may be task-dependent.
- As the complexity of the classification task increased, the trade-off between annotation accuracy and cost became more pronounced. While GPT-only methods always offered the lowest cost, their performance gap with human annotations widened in more complex tasks, necessitating the use of hybrid or human-centric approaches to maintain high accuracy.
- As dataset portions increased from 10% to 50%, we observed that the performance gap between different methods often narrowed, while cost differences became more pronounced. This trend emphasizes the importance of considering scalability when choosing annotation strategies for larger datasets.

These findings collectively demonstrate the potential of integrating LLMs into Active Learning frameworks for text classification tasks. They highlight the need for tailored approaches based on task complexity and desired performance levels, and underscore the potential for significant cost savings through strategic use of LLM-based annotations.

Our results suggest that for simpler tasks like sentiment analysis, GPT-based approaches with minimal human intervention can offer an optimal balance of cost and accuracy. For more complex tasks, such as fake news detection or multi-class classification, hybrid approaches leveraging high-confidence LLM annotations with selective human oversight appear to be most effective.

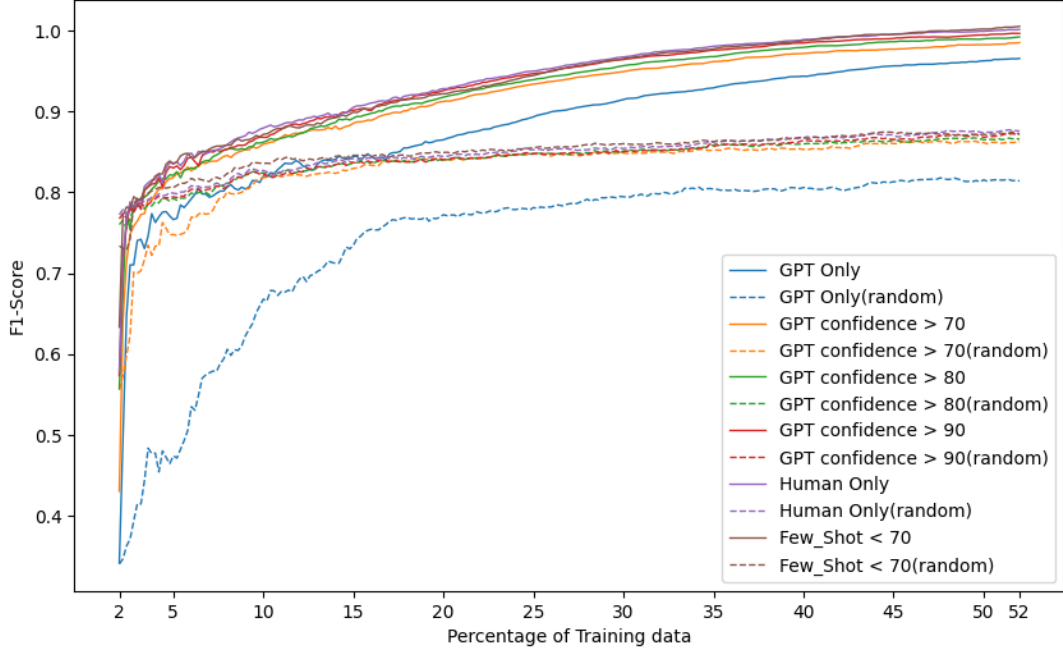


Figure 5.1: Evaluation of F1 scores of the IMDB Dataset across different scenarios

Moving forward, we’ll explore how these quantitative findings are reflected in our visualized results, and examine the implications of our novel proxy validation approach and role design experiments.

5.2 Graphical Analysis of Results

To better illustrate the trends and patterns in our data, we’ve created several visualizations that highlight key aspects of our findings across the three datasets.

Our visualized results provide a compelling illustration of the performance and cost dynamics across different annotation methods for our three datasets. Beginning with the IMDB dataset, Figure 5.1 showcases the evolution of F1 scores as the training data portion increases from 2% to 52%. This visualization strikingly demonstrates the superiority of our active learning approach over random sampling across all annotation methods. The graph reveals a consistent upward trajectory for all methods, with the GPT-based and hybrid approaches closely trailing the performance of human-only an-

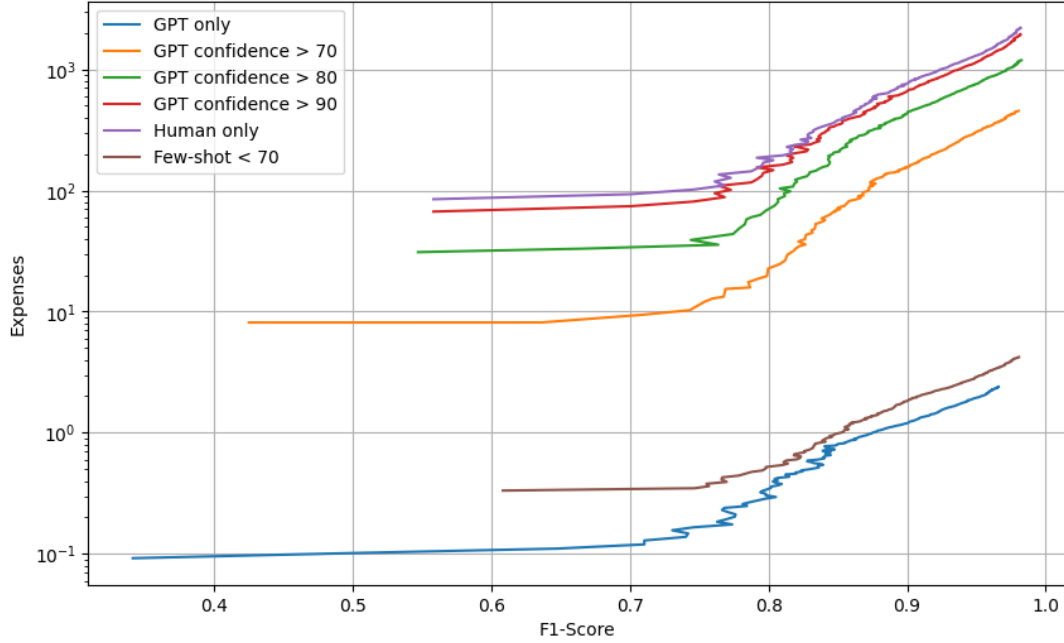


Figure 5.2: Cost per F1 Score Analysis During Iterative Training Data Increments in the IMDB Dataset.

notations. Notably, the few-shot learning method for low-confidence data points (below 70%) shows remarkable efficiency, achieving performance comparable to more expensive methods at a fraction of the cost.

The cost implications of these different annotation setups for the IMDB dataset are vividly captured in Figure 5.2, which uses a logarithmic scale to highlight the stark cost disparities. This representation underscores the significant cost-saving potential of GPT-based methods. The GPT-only and few-shot approaches maintain relatively low costs even as F1 scores improve with increasing data portions. In contrast, human-only and high-confidence hybrid methods exhibit a steeper cost increase, albeit with marginally better performance. This trend suggests that for sentiment analysis tasks, where LLMs perform reasonably well, a GPT-based approach with selective human intervention could offer an optimal balance of cost and accuracy.

Shifting our focus to the Fake News dataset, Figure 5.3 paints a similar picture of performance evolution, albeit with some notable differences. The gap between human-

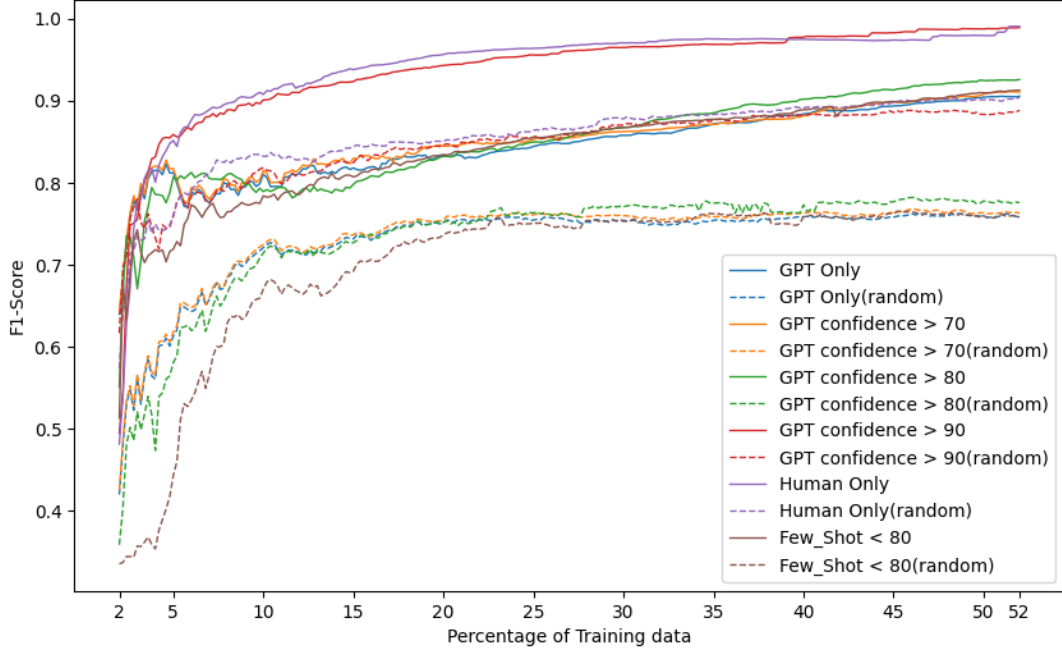


Figure 5.3: Evaluation of F1 scores of the Fake NEWS Dataset across different scenarios

only annotations and other methods is more pronounced in this dataset, particularly in the early stages of training. However, as the data portion increases, we observe a convergence in performance across methods, with hybrid approaches, especially those with high confidence thresholds, closely approximating human-level accuracy. This trend underscores the potential of LLM integration in more nuanced classification tasks, where initial performance may lag but can be significantly improved with larger training sets.

The cost dynamics for the Fake News dataset, illustrated in Figure 5.4, reveal a more complex landscape. While GPT-only methods maintain their cost advantage, the performance benefits of hybrid approaches become more apparent. The GPT conf ≥ 90 method, in particular, shows a favorable cost-to-performance ratio, approaching human-level accuracy at a fraction of the cost. This finding highlights the potential of high-confidence LLM annotations in tasks where the complexity necessitates more selective use of human annotation.

Lastly, our visualization of the Movie Genres dataset results in Figure 5.5 presents the most challenging scenario. The multi-class nature of this task is reflected in the

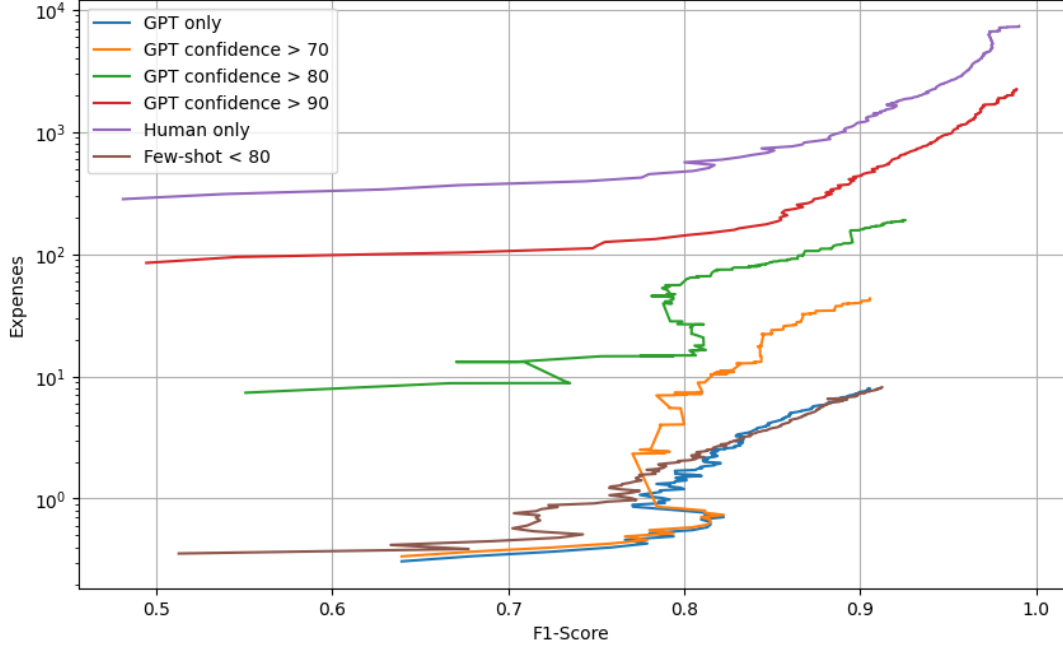


Figure 5.4: Cost per F1 Score Analysis During Iterative Training Data Increments in the Fake NEWS Dataset.

generally lower F1 scores across all methods compared to the binary classification tasks. The performance gap between human-only annotations and LLM-based methods is more pronounced, especially in the early stages of training. However, as the data portion increases, we observe a gradual convergence, with hybrid methods, particularly those with high confidence thresholds, showing significant improvements.

The cost analysis for the Movie Genres dataset, depicted in Figure 5.6, further emphasizes the trade-offs between performance and cost in complex classification tasks. While GPT-only and low-confidence hybrid methods maintain low costs, their performance improvement plateaus quickly. The GPT conf ≥ 90 method emerges as a promising middle ground, achieving a balance between cost-effectiveness and performance that approaches human-level accuracy. This underscores the potential of high-confidence LLM annotations in multi-class classification tasks, where the complexity demands a more nuanced approach to balancing automated and human annotations.

These visualized results collectively reinforce our quantitative findings, providing a

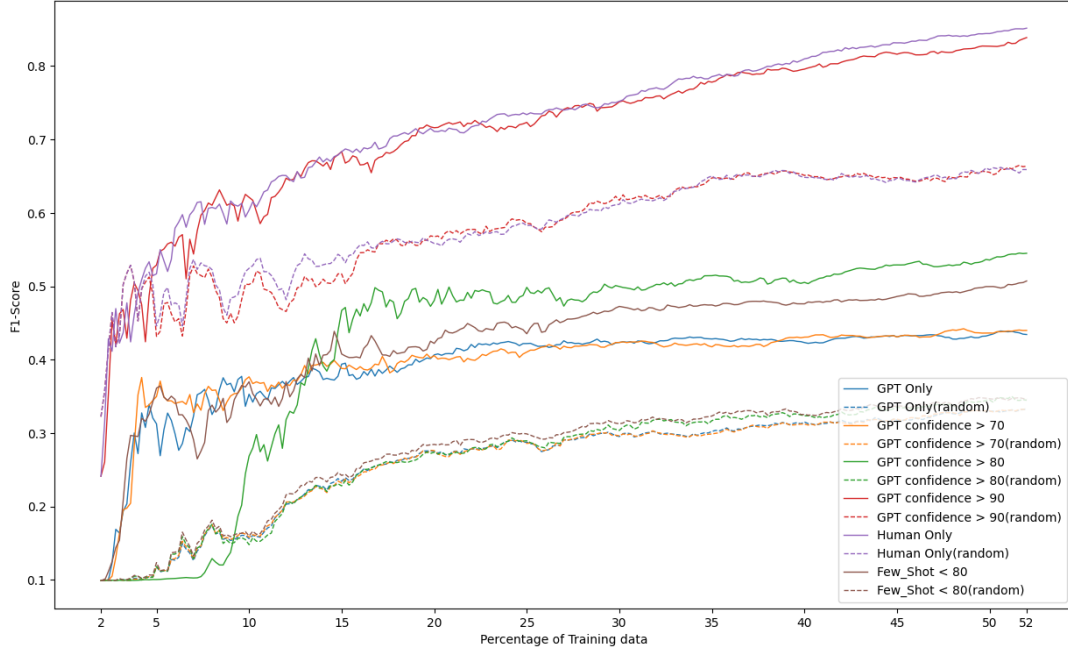


Figure 5.5: Evaluation of F1 scores of the Movie Genres Dataset across different scenarios

clear and intuitive representation of the performance trajectories and cost dynamics across different annotation methods and datasets. They highlight the potential of LLM integration in active learning frameworks, while also illustrating the need for tailored approaches based on task complexity and desired performance levels.

At the conclusion of our visualized results analysis, it's crucial to examine the cost-effectiveness of different annotation methods and discuss potential stopping criteria for the active learning process. Figures 5.8, 5.9, and 5.7 provide valuable insights into the economic implications of improving model performance across our three datasets.

These figures illustrate the cost required to achieve each 1% improvement in F1 score as the percentage of training data increases. This metric offers a nuanced view of the diminishing returns in model performance improvement relative to annotation costs. As we progress through iterations of data labeling, we generally observe that achieving performance improvements becomes increasingly costly, represented by upward trends in these graphs.

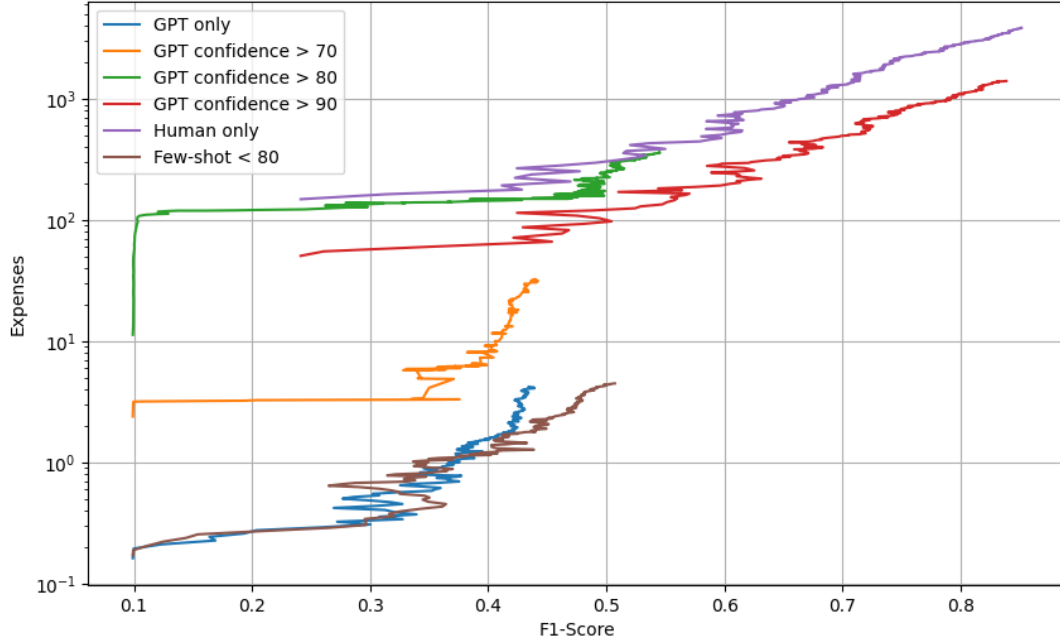


Figure 5.6: Cost per F1 Score Analysis During Iterative Training Data Increments in the Movie Genres Dataset.

For instance, in the IMDB dataset (Figure 5.7), we notice that the human-only annotation method shows a relatively steady line after approximately 42% of data labeling. This plateau indicates that from 42% to 52% of labeled data, there was no significant improvement (less than 1%) in the F1 score, despite a 10% increase in labeled data. This observation suggests a potential stopping point for the annotation process, as further labeling would likely yield diminishing returns relative to the cost.

Similarly, in the Movie Genres dataset (Figure 5.9), we can observe varying patterns of cost-effectiveness across different annotation methods. The GPT-only and low-confidence hybrid methods maintain relatively low costs but show limited improvement in later stages. In contrast, the high-confidence hybrid methods (e.g., GPT conf ≥ 90) demonstrate a more favorable cost-to-performance ratio, approaching human-level accuracy at a fraction of the cost.

The Fake News dataset results (Figure 5.8) further reinforce these trends, showing how different annotation strategies can significantly impact the cost-effectiveness of the

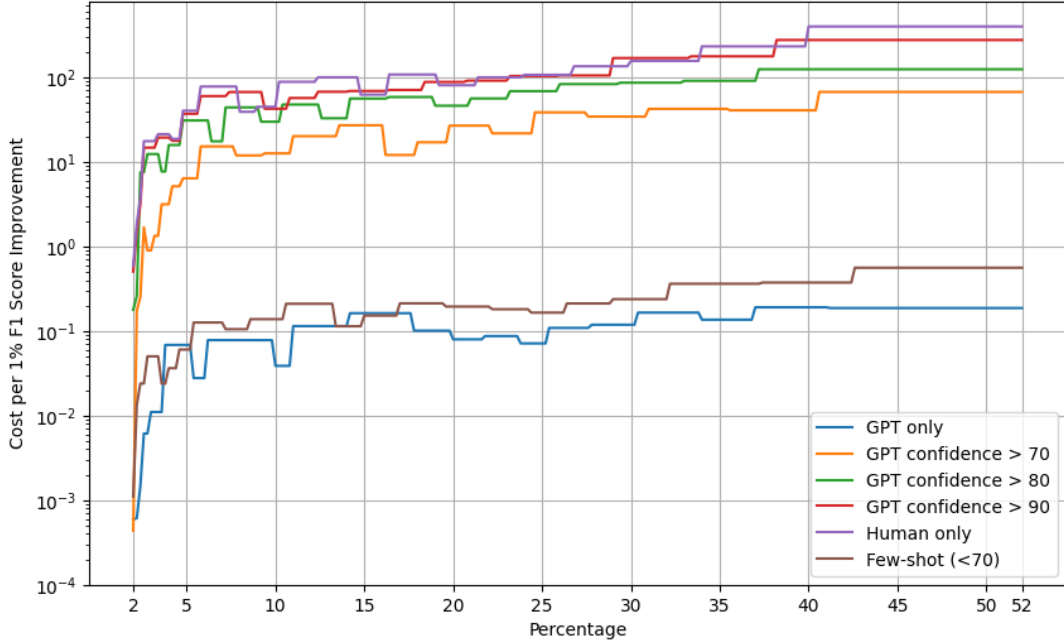


Figure 5.7: Cost-effectiveness analysis for the IMDB dataset

learning process. These visualizations allow us to establish data-driven stopping criteria for the active learning process. We can define stopping points based on:

1. **Performance Plateau:** When we observe a horizontal line extending over a specific portion of the graph, it indicates a lack of significant improvement despite increased labeling. This can serve as a natural stopping point.
2. **Cost Threshold:** We can set a maximum acceptable cost for each 1% improvement in F1 score. As the cost per improvement increases (visible on the Y-axis), we can determine a threshold beyond which further annotation is deemed economically unfeasible.
3. **Improvement Rate:** If the improvement in F1 score falls below a certain threshold (e.g., less than 1% improvement over a 10% increase in labeled data), we might consider halting the process.
4. **Budget Constraints:** Based on these graphs, we can estimate the budget required to reach certain performance levels and stop when we hit predefined budget limits.

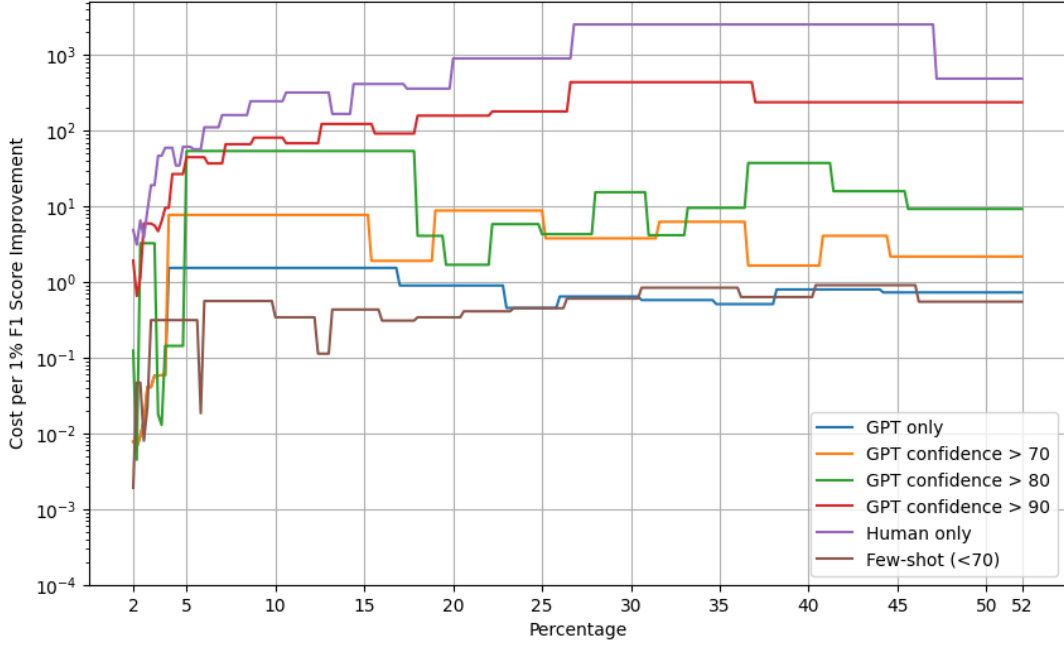


Figure 5.8: Cost-effectiveness analysis for the Fake News dataset

It's worth noting that while the overall trend shows increasing costs for improvements, there are some fluctuations where the cost for a 1% improvement may temporarily decrease. However, these fluctuations are less common in the earlier iterations and become more pronounced as we progress further in the labeling process.

These cost-effectiveness analyses provide valuable insights for optimizing the annotation process. They allow us to balance the trade-off between model performance and annotation costs, ensuring efficient resource allocation in real-world applications of active learning for text classification tasks. By carefully considering these metrics and visualizations, researchers and practitioners can make informed decisions about when to terminate the active learning process, maximizing performance gains while minimizing unnecessary expenditure on data annotation.

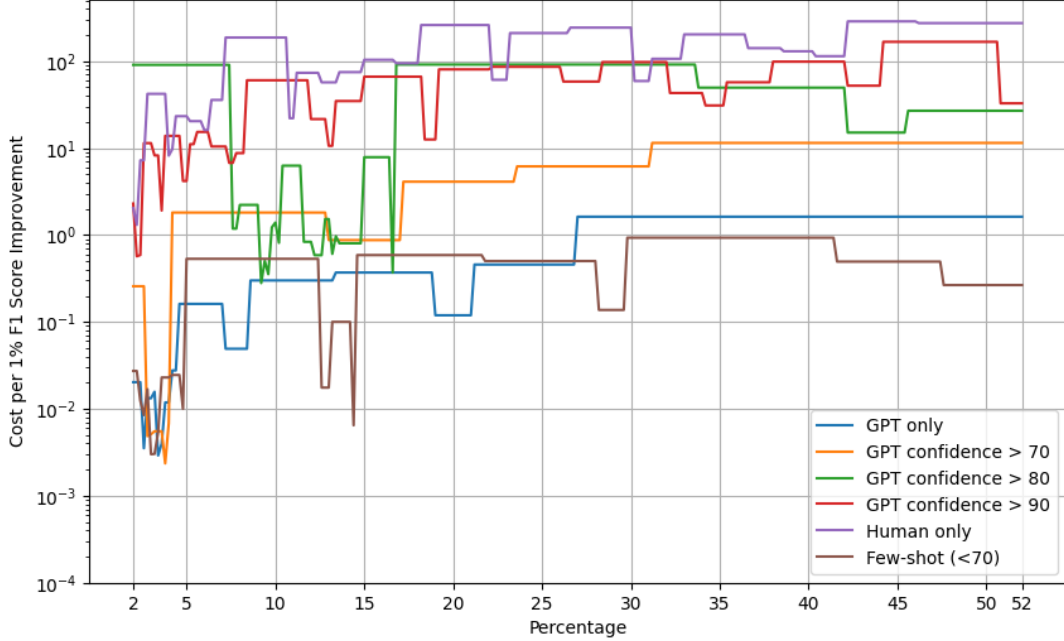


Figure 5.9: Cost-effectiveness analysis for the Movie Genres dataset

5.3 Proxy Validation Results

A key innovation in our study was the implementation of a proxy validation approach to estimate model performance and overall pool quality. Table 5.4 presents a comprehensive analysis of this approach across our three datasets and various annotation methods. The results demonstrate the effectiveness and reliability of our proxy validation strategy as an indicator of overall pool quality.

For the IMDB dataset, we observe consistently high correlations across all variations, with Pearson and Spearman correlation coefficients exceeding 0.98 in most cases. The GPT-only method shows particularly strong performance, with the highest Pearson (0.9928) and Spearman (0.9959) correlations, as well as the highest R-squared value (0.9514). This indicates that our proxy validation approach is especially effective for sentiment analysis tasks, where GPT-3.5 performs well. The low Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) values, ranging from 0.0139 to 0.0180, further support the accuracy of our proxy validation method.

Dataset	Variation	Pearson	Spearman	MAE	RMSE	R-squared
IMDB	GPT only	0.9928	0.9959	0.0155	0.0167	0.9514
	GPT conf > 70	0.9898	0.9873	0.0160	0.0171	0.9347
	GPT conf > 80	0.9927	0.9918	0.0139	0.0153	0.9415
	GPT conf > 90	0.9929	0.9883	0.0161	0.0174	0.9216
	Human only	0.9885	0.9797	0.0151	0.0180	0.9157
	Few-shot < 70	0.9838	0.9948	0.0149	0.0171	0.9197
Fake News	GPT only	0.9785	0.9817	0.0074	0.0092	0.9503
	GPT conf > 70	0.9638	0.9741	0.0092	0.0117	0.9154
	GPT conf > 80	0.9914	0.9870	0.0182	0.0197	0.8742
	GPT conf > 90	0.9840	0.9812	0.0157	0.0179	0.9136
	Human only	0.9910	0.9850	0.0210	0.0225	0.8730
	Few-shot < 70	0.9877	0.9840	0.0221	0.0241	0.8360
Movie Genres	GPT only	0.9351	0.5017	0.0215	0.0249	0.8253
	GPT conf > 70	0.9603	0.6620	0.0136	0.0168	0.9142
	GPT conf > 80	0.9900	0.8603	0.0212	0.0264	0.9673
	GPT conf > 90	0.9699	0.9622	0.0426	0.0481	0.7927
	Human only	0.9735	0.9621	0.0258	0.0321	0.9101
	Few-shot < 70	0.9720	0.8832	0.0151	0.0180	0.9444

Table 5.4: Proxy Validation Results: This table presents the key metrics for proxy validation across three datasets (IMDB, Fake News, and Movie Genres) and various confidence thresholds.

The Fake News dataset results reveal strong correlations across most variations, with Pearson coefficients above 0.96 and Spearman coefficients generally above 0.97. Notably, the GPT conf > 80 method achieves the highest Pearson correlation (0.9914) and a very high Spearman correlation (0.9870). The human-only method shows an interesting discrepancy between its high Pearson correlation (0.9910) and lower Spearman correlation (0.6796), suggesting that while the overall trend is captured well, the ranking of individual instances may be less consistent. The generally low MAE and RMSE values (between 0.0074 and 0.0241) indicate high accuracy in our proxy validation estimates.

For the more complex Movie Genres dataset, we observe more variability in the results, reflecting the increased challenge of multi-class classification. The GPT conf > 80 method shows the strongest performance, with the highest Pearson correlation (0.9900)

and R-squared value (0.9673). The GPT-only method demonstrates lower correlations, particularly in terms of Spearman correlation (0.5017), suggesting that for more complex tasks, confidence-based hybrid approaches may provide more reliable proxy validation. The MAE and RMSE values are generally higher for this dataset, ranging from 0.0136 to 0.0481, indicating increased difficulty in accurately estimating performance for multi-class problems.

These results collectively underscore the robustness and reliability of our proxy validation approach as an indicator of overall pool quality. The consistently high correlations and low error rates across diverse datasets and annotation methods demonstrate that our proxy validation set effectively mirrors the performance trends of the larger data pool. This validates our approach as a valuable tool for estimating model performance and guiding decision-making in active learning scenarios, particularly when full labeled datasets are not available for validation.

The effectiveness of our proxy validation method has significant implications for practical applications of active learning in text classification tasks. It provides a reliable means of monitoring model performance and data quality throughout the learning process, potentially reducing the need for extensive human validation and allowing for more efficient allocation of annotation resources.

5.4 Role Design Experiments

In the context of role design in prompts, our experiments with various LLMs revealed intriguing patterns, as shown in Table 5.5. The few-shot system, user, and assistant prompt (FewSUA) configuration consistently exhibited superior F1 scores across different datasets and models. This finding underscores the importance of incorporating clear role distinctions and examples in enhancing model performance. Interestingly, for the Llama models, using few-shot user (FewU) and few-shot system and user (FewSU) prompts

Dataset	LLM	ZeroU		ZeroSU		FewU		FewSU		FewSUA	
		Str. Acc.	F1 Score	Str. Acc.	F1 Score	Str. Acc.	F1 Score	Str. Acc.	F1 Score	Str. Acc.	F1 Score
commonsense_qa	GPT-3.5	24	68	60	68	20	69	55	68	100	73
	GPT-4o	100	77	100	80	99	79	100	82	100	83
	Llama2-7b	0	19	0	19	0	18	0	19	67	9
	Llama2-13b	0	33	0	36	0	36	0	36	99	28
ai2_arc	GPT-3.5	73	76	39	80	60	80	87	78	99	85
	GPT-4o	99	96	100	96	100	95	100	96	100	97
	Llama2-7b	0	36	0	39	0	26	0	25	9	40
	Llama2-13b	0	50	0	52	0	39	0	37	77	48
wiki_movie_plots	GPT-3.5	99	76	99	79	99	77	99	76	99	77
	GPT-4o	100	80	100	81	100	81	100	82	100	84
	Llama2-7b	0	75	0	75	0	68	0	70	25	74
	Llama2-13b	0	73	0	75	0	72	0	75	64	85
IMDB_reviews	GPT-3.5	100	94	100	93	100	93	100	67	100	94
	GPT-4o	100	95	100	95	100	96	100	92	100	97
	Llama2-7b	1	87	1	87	0	62	0	60	18	85
	Llama2-13b	3	91	3	90	0	64	0	67	82	93

Table 5.5: Performance of language models with various prompt designs across datasets. Shows Structural Accuracy (Str. Acc.) and F1 scores for ZeroU, ZeroSU, FewU, FewSU, and FewSUA methods.

led to a noticeable decrease in F1 scores, suggesting that embedding few-shot examples within user prompts may cause these models to produce more incorrect answers and hallucinations.

The structural accuracy of GPT models was generally high, with both GPT-3.5 and GPT-4 adhering to the desired output structure in most cases. In contrast, the Llama models frequently generated outputs that didn’t follow the instructions precisely, which is understandable given their smaller size. A standout finding is that the FewSUA prompt configuration helped the Llama models generate outputs in the desired structure, making it potentially useful for applications like chatbots where maintaining a specific output format is crucial.

In tasks such as movie genre classification and sentiment analysis, the Llama2-13b model showed results comparable to the GPT models, demonstrating its capability in

certain NLP tasks despite its smaller size. However, in question-answering tasks, the performance of the GPT models was significantly better and not comparable to that of the Llama models. This performance disparity highlights the importance of model selection based on the specific requirements of the task at hand.

Overall, our experiments indicate that larger models generally achieve better F1 scores and exhibit improved structural adherence. GPT-4 consistently outperformed GPT-3.5, and Llama2-13b showed superior results compared to Llama2-7b. This trend suggests that model size plays a crucial role in both accuracy and the ability to follow structural guidelines in outputs.

These findings collectively paint a comprehensive picture of the potential for integrating LLMs into Active Learning frameworks for text classification tasks. They highlight the intricate interplay between model confidence, annotation costs, and classification accuracy, providing valuable insights for optimizing annotation strategies in various NLP applications.

The impact of role design in prompts on model performance across different datasets and LLMs, as illustrated in Table 5.5, provides valuable insights into optimizing interactions with these models. For the commonsense_qa dataset, we observe that GPT-4o consistently outperformed GPT-3.5 across all prompt designs, with the FewSUA configuration yielding the highest F1 score of 83% for GPT-4o. Interestingly, the Llama models struggled with this task, with Llama2-13b showing slightly better performance than Llama2-7b, but still significantly lower than the GPT models.

In the ai2_arc dataset, which focuses on grade-school level science questions, GPT-4o demonstrated exceptional performance, achieving F1 scores above 95% across all prompt designs. The FewSUA configuration again proved most effective, yielding a peak F1 score of 97%. GPT-3.5 also performed well, with the FewSUA design boosting its F1 score to 85%. The Llama models showed improved performance on this dataset compared to commonsense_qa, but still lagged behind the GPT models.

For the Movie Genres dataset, we see a more level playing field across models. GPT-4o maintained its superior performance, with the FewSUA configuration yielding the highest F1 score of 84%. Notably, Llama2-13b achieved an impressive 85% F1 score with the FewSUA design, outperforming even GPT-3.5 on this task. This result suggests that for certain domain-specific tasks, fine-tuned smaller models can compete with larger, more general models.

The IMDB dataset showcased high performance across all models, with even the Llama models achieving F1 scores above 85% in most configurations. GPT-4o again led the pack with a 97% F1 score using the FewSUA design, closely followed by GPT-3.5 at 94%. The strong performance of Llama models on this task, particularly Llama2-13b achieving a 93% F1 score with FewSUA, indicates that sentiment analysis may be a task where smaller, efficiently designed models can approach the performance of larger models.

A crucial aspect revealed in Table 5.5 is the structural accuracy of model outputs. GPT models consistently demonstrated high structural accuracy across all datasets and prompt designs, often achieving 100%. This indicates their strong ability to adhere to specified output formats, a critical feature for many practical applications. In contrast, Llama models struggled with structural accuracy in most cases, particularly with zero-shot and few-shot user-only prompts. However, the FewSUA configuration significantly improved the structural accuracy of Llama models, especially for Llama2-13b, suggesting that clear role delineation in prompts can enhance the model’s ability to follow instructions.

These results highlight several key trends in LLM performance and prompt design. Firstly, the consistent superiority of the FewSUA configuration across models and tasks underscores the importance of clear role distinction and example provision in prompts. This design appears to help models, especially smaller ones like Llama, to better understand and adhere to task requirements.

Secondly, the performance gap between larger models (GPT-4o, GPT-3.5) and smaller models (Llama2-13b, Llama2-7b) varies depending on the task. While GPT models generally outperform Llama models, the margin of difference is task-dependent. In some cases, like movie plot genre classification and sentiment analysis, Llama2-13b approaches or even matches the performance of GPT-3.5, suggesting that for certain domain-specific tasks, smaller models can be competitive alternatives.

Thirdly, the structural accuracy results emphasize the importance of prompt design in eliciting desired output formats. The dramatic improvement in Llama models’ structural accuracy with the FewSUA configuration indicates that careful prompt engineering can partially compensate for limitations in model size or training.

These findings have significant implications for the practical application of LLMs in various NLP tasks. They suggest that while larger models like GPT-4o generally offer superior performance, there are scenarios where smaller, more efficiently deployable models can be effectively utilized, especially when coupled with optimal prompt designs. The choice of model and prompt configuration should be guided by the specific requirements of the task, balancing factors such as performance, computational resources, and the need for structural adherence in outputs.

Moreover, the consistent improvement seen with the FewSUA configuration across models and tasks points to a promising direction for standardizing prompt designs in LLM applications. This could lead to more efficient and effective utilization of LLMs in diverse NLP tasks, potentially reducing the need for task-specific fine-tuning and improving the generalizability of these models across different applications.

Chapter 6

Conclusion and Future Works

6.1 Conclusion

This thesis presents a comprehensive investigation into the integration of Large Language Models (LLMs) within Active Learning frameworks for text classification tasks. Our research demonstrates the significant potential of this approach to enhance both the efficiency and accuracy of text classification while substantially reducing annotation costs. Through a series of experiments across diverse datasets and task complexities, we have uncovered several key insights that contribute to the growing body of knowledge in this field.

Firstly, our results consistently show that LLM-based annotation methods, particularly when combined with strategic human oversight, can achieve performance levels comparable to human-only annotations at a fraction of the cost. This was especially evident in the IMDB sentiment analysis task, where GPT-3.5 annotations nearly matched human-level accuracy while reducing costs by over 99%. Even in more complex tasks like fake news detection and multi-class movie genre classification, hybrid approaches leveraging high-confidence LLM annotations with selective human intervention proved highly effective.

Secondly, the implementation of our novel proxy validation approach has demonstrated its reliability as an indicator of overall pool quality. The consistently high correlations and low error rates across diverse datasets and annotation methods validate this approach as a valuable tool for estimating model performance in active learning scenarios, particularly when full labeled datasets are not available for validation. This innovation addresses a critical challenge in active learning, providing a robust method for monitoring model performance and data quality throughout the learning process.

Thirdly, our exploration of role design in prompts has revealed the significant impact of prompt engineering on LLM performance. The Few-shot System, User, and Assistant (FewSUA) configuration consistently outperformed other designs across different models and tasks, highlighting the importance of clear role distinctions and example provision in enhancing model performance. This finding offers valuable guidance for optimizing LLM interactions in various NLP applications.

Furthermore, our research underscores the importance of tailoring annotation strategies to specific task requirements and complexities. While GPT-only methods offered substantial cost savings across all tasks, their effectiveness varied with task complexity. For simpler tasks like sentiment analysis, GPT-based approaches with minimal human intervention provided an optimal balance of cost and accuracy. However, for more complex tasks such as multi-class classification, hybrid approaches leveraging high-confidence LLM annotations with selective human oversight emerged as the most effective strategy.

In conclusion, this research demonstrates the transformative potential of integrating LLMs into Active Learning frameworks for text classification tasks. By strategically combining the strengths of LLMs, human expertise, and carefully designed active learning strategies, we can significantly enhance the efficiency and cost-effectiveness of text classification while maintaining high levels of accuracy. These findings have broad implications for various domains requiring large-scale text analysis, offering a pathway to more efficient and accessible NLP solutions.

6.2 Future Works

Our research contributes to understanding the integration of LLMs into Active Learning for text classification, suggesting several potential avenues for future research.

Firstly, with the continual emergence of more powerful LLMs, future studies could explore their performance within our proposed framework. This includes evaluating the capabilities of various models, including open-source alternatives. Additionally, examining the impact of fine-tuning LLMs on domain-specific data before their integration into the Active Learning framework could enhance performance for specialized tasks or niche domains.

Another promising area is the development of adaptive algorithms that dynamically adjust confidence thresholds based on model performance and task complexity, optimizing the balance between LLM and human annotations. Extending our approach to incorporate multimodal data, such as text combined with images or audio, could also broaden the applicability of this framework to diverse real-world scenarios.

Optimizing the computational efficiency of LLM-integrated Active Learning, particularly for large datasets, could improve scalability. Investigating the effectiveness of this approach in cross-lingual and multilingual settings might open new possibilities for global-scale text classification tasks. Additionally, exploring how Active Learning principles could be applied to the fine-tuning process of LLMs themselves could lead to more efficient and targeted model improvements.

In addition, future work could explore enhancements to uncertainty measures in multi-class scenarios. In cases where probability scores are closely distributed among some classes while significantly lower for others, conventional uncertainty sampling may not fully capture the true ambiguity of these instances. Employing variance-based uncertainty measures could address this limitation more effectively, allowing for a refined selection of informative samples.

Finally, future research in role-based prompt engineering could explore several promis-

ing directions. These include investigating the transferability of effective role designs across diverse domains and tasks and examining the impact of culturally diverse role designs on model performance. A key area of focus should be the integration of dynamic role adaptation within the active learning loop, which could potentially enhance the synergy between LLM performance and the active learning process. This approach could involve developing algorithms that automatically adjust role structures based on model performance and task complexity.

These future directions have the potential to further advance text classification and expand the applicability of LLM-integrated Active Learning across various domains and use cases. As LLMs continue to evolve, their integration with Active Learning frameworks presents an exciting frontier to enhance the efficiency, accuracy, and accessibility of text analysis tasks.

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Chapter 7

Appendix

7.1 Prompt Design Phase 1

To illustrate the type of prompt used for GPT-3.5 annotation, consider the example shown in Figure 7.1 for the IMDB dataset.

```
1 prompt = f"""
2     What is the sentiment of the following movie review,
3     and how confident you are about this 'sentiment',
4     which is delimited with triple backticks?
5
6     Give your answer as a single word, either 'positive'
7     or 'negative' and a single percentage in JSON format
8     delimited with space.
9
10    Review text: '{X[i]}'
    """
```

Figure 7.1: Example of a GPT-3.5 prompt for sentiment analysis in movie reviews, formatted to request output in JSON format.

This prompt was designed to be straightforward, directing GPT-3.5 to classify a given movie review as either Positive or Negative. The simplicity of the prompt ensures clarity in the task, allowing GPT-3.5 to focus solely on the sentiment analysis of the provided review text.

The example provided in Figure 7.2 shows a few-shot prompt used in the study for the Movie Genres dataset.

```
1 prompt = f"""
2 Determine the genre of the movie based on the following plot:
3
4 For the plot provided, classify its genre as a single word (
5     without other marks or words like 'genre:'), either "comedy
6     ", "action", "drama", or "horror".
7
8 Use the few-shot learning examples below to improve your
9     prediction:
10
11 Few-shot Examples:
12 '{newsetX.iloc[0]}' genre:{newsetY.iloc[0]}
13 '{newsetX.iloc[1]}' genre:{newsetY.iloc[1]}
14 '{newsetX.iloc[2]}' genre:{newsetY.iloc[2]}
15 Movie plot:
16 '{newsetX.iloc[i]}'
17 """
```

Figure 7.2: Example of a GPT-3.5 few-shot learning prompt for Movie Genres classification.

This prompt provides GPT-3.5 with a few examples to illustrate the task, followed by a new description for classification. The structure of the prompt is key in ‘teaching’ the model and the nature of the task, using just a few examples, enabling it to apply this understanding to new, unseen descriptions.

7.2 Prompt Examples Phase 2

The prompt designs include ZeroU (Zero-shot User-only), ZeroSU (Zero-shot System and User), FewU (Few-shot User-only), FewSU (Few-shot System and User), and FewSUA (Few-shot System, User, and Assistant). Each design isolates the effect of role-based structuring on model performance.

While the specific example in Figure 7.3 is for the `wiki_movie_plots` dataset, similar prompts are used for other datasets with minor changes in the desired output, such as

sentiment for IMDB reviews or answers for commonsense_qa and ai2_arc. By splitting the prompts into distinct roles, the study observes the impact on the ability of models to generate accurate and structured responses.

Figures 7.4, 7.5, and 7.6 present examples of structured and unstructured responses across different datasets, including sentiment analysis, genre classification, and question-answering tasks. Structured responses are critical for consistency in evaluation, requiring specific formats such as single words or letters.

The post-processing technique mentioned in the text is essential for accurately extracting these labels from the model outputs, highlighted by the blue-colored text in the examples. However, there are instances where the extraction process makes mistakes, as shown in Figure 7.5. For example, in the GPT-4o (commonsense_qa) example, the post-processing incorrectly extracted ‘A’ as the answer, although the output indicated ‘C’. This highlights the challenges in achieving precise output structures and the importance of improving post-processing techniques to avoid such errors.

The appendix emphasizes the importance of clear role distinctions in prompt design, particularly the FewSUA configuration, which consistently improves both F1 scores and structural adherence. The detailed examples and discussions provide valuable insights for developing effective prompt designs, highlighting the crucial role of post-processing in maintaining output accuracy.

ZeroU prompt:

role: “**user**”, content: Determine the genre of the movie based on the provided plot:

For the plot provided, classify its genre as a single word (without other marks or words like ‘genre:’), either “comedy”, “action”, “drama”, or “horror”.

Plot[i]

ZeroSU prompt:

role: “**system**”, content: Determine the genre of the movie based on the provided plot.

For the plot provided, classify its genre as a single word (without other marks or words like ‘genre:’), either “comedy”, “action”, “drama”, or “horror”.

role: “**user**”, content: Plot[i]

FewU prompt:

role: “**user**”, content: Determine the genre of the movie based on the provided plot:

For the plot provided, classify its genre as a single word (without other marks or words like ‘genre:’), either “comedy”, “action”, “drama”, or “horror”.

Examples:

Plot[1] - Genre[1]

Plot[2] - Genre[2]

Plot[3] - Genre[3]

...

Plot[i]

FewSU prompt:

role: “**system**”, content: Determine the genre of the movie based on the provided plot.

For the plot provided, classify its genre as a single word (without other marks or words like ‘genre:’), either “comedy”, “action”, “drama”, or “horror”.

role: “**user**”, content: Examples:

Plot[1] - Genre[1]

Plot[2] - Genre[2]

Plot[3] - Genre[3]

...

Plot[i]

FewSUA prompt:

role: “**system**”, content: Determine the genre of the movie based on the provided plot.

For the plot provided, classify its genre as a single word (without other marks or words like ‘genre:’), either “comedy”, “action”, “drama”, or “horror”.

role: “**user**”, content: Plot[1]

role: “**assistant**”, content: Genre[1]

role: “**user**”, content: Plot[2]

role: “**assistant**”, content: Genre[2]

role: “**user**”, content: Plot[3]

role: “**assistant**”, content: Genre[3]

role: “**user**”, content: Plot[i]

Figure 7.3: This figure illustrates the various prompt designs utilized to feed all models.

Sentiment (IMDB reviews):
positive / negative (single word)

Genre (wiki_movie_plots):
comedy, drama, action, horror (single word)

QA (ai2_arc):
A to E (single letter)

QA (commonsense_qa):
A to D (single letter)

Figure 7.4: This figure illustrates the various structured responses for different datasets used to evaluate the models.

GPT-4o (IMDB reviews):
neutral

GPT-4o (commonsense_qa):
Greg’s performance was mediocre. What might his habits be like? A: distinguished
B: unique C: peculiar D: exceptional E: outstanding
The answer is: C

GPT-3.5 (IMDB reviews):
science fiction

GPT-3.5 (commonsense_qa):
D: to remove toxic material from the stomach before it is absorbed

Figure 7.5: This figure shows examples of unstructured responses from GPT models. The blue-colored text highlights the extracted labels in the examples.

Llama2-7b (commonsense_qa):

Based on the question provided, the answer is:

A: halving the distance between them

Therefore, the answer is “A”.

Llama2-7b (IMDB reviews):

The sentiment of the review is **negative**. The reviewer expresses disappointment with the sequel compared to the original, citing a lack of fun and too much time spent on plot development. They also mention that Tim Thomerson’s wisecracking is toned down, and the performances are adequate but the script lets them down. The action is described as routine and the plot is only mildly interesting, which suggests that the reviewer was not impressed with the film’s action sequences or story. The reviewer also states that they need lots of silly laughs to stay entertained during a Trancers movie, but the film fails to deliver on this front. Overall, the review is negative and suggests that the reviewer did not enjoy the sequel as much as the original.

Llama2-7b (wiki_movie_plots):

Based on the provided plot, I would classify the movie as a **comedy**. The plot involves a group of irresponsible and wealthy individuals who die in an accident and become ghosts, with the goal of freeing their friend from a regimented lifestyle in order to enter heaven. The movie appears to have a lighthearted and comedic tone, with elements of slapstick humor, witty dialogue, and farcical situations. The characters’ antics and misadventures are likely to provide most of the comedic elements in the movie.

Llama2-13b (wiki_movie_plots):

Sure! Based on the plot provided, the genre of the movie would be:

Comedy.

The plot involves a man trying to relive his youth and pursue a beautiful woman, with his friends, in a series of comedic and awkward situations. The tone of the plot is lighthearted and humorous, indicating that the movie would be a comedy.

Llama2-13b (IMDB reviews):

Based on the review, I would rate the movie 3 out of 10. Here’s a breakdown of the review:

- * The reviewer can’t knock the movie too hard because it’s clear that the filmmakers were trying to make it bad or campy.
- * Some of the parts that were intended to be funny actually are, but only for their cheesiness.
- * The reviewer notes that the movie goes over the top near the end with a time warp and sexual innuendoes.
- * The reviewer is surprised that the movie got a PG rating, implying that it’s not suitable for all audiences.
- * The reviewer concludes that it’s a fun way to waste a couple of hours, but it’s not on the same level as Star Wars or Trek.

Llama2-13b (ai2_arc):

Sure! Based on the question provided, the answer is:

D : wildfire

Figure 7.6: This figure shows examples of unstructured responses from Llama2 models. The blue-colored text highlights the extracted labels in the examples.